सत्यमेव जयते



भारत सरकार/ Government of India

विद्युत मंत्रालय/ Ministry of Power

केन्द्रीय विद्युत प्राधिकरण/ Central Electricity Authority

विद्युत प्रणाली अभियांत्रिकी एवं प्रौद्योगिकी विकास प्रभाग

Power System Engineering & Technology Development Division

Date: 06.08.2025

सेवा में,

<As per attached list>

विषय: Minutes of the meeting to Deliberate on Outage of 400 kV Transmission Lines of Mumbai Area– reg.

महोदया/ महोदय,

A meeting was held under the chairmanship of Member (Power Systems), CEA on 04.08.2025 at 11:30 hrs at through VC to deliberate on tripping pf 400 kV Transmission Lines feeding Mumbai Area that occurred on 30-31 July, 2025.

The minutes of the meeting is attached for your information and necessary action.

Encl.: As above

भवदीय,

Signed by Bhanwar Singh Meena
Date: 06-08-2025 14:49:32

(भंवर सिंह मीना /Bhanwar Singh Meena)

निदेशक/ Director

To,

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2. **Chief Engineer, Grid Management Division, CEA Email: gmcea@nic.in, brieflee.lyngkhoi@gov.in**
3. **Executive Director, WRLDC Email: mehendale@grid-india.in, satyendra@grid-india.in, tmuthukumar@grid-india.in, madhukar@grid-india.in, so2wrldc@grid-india.in, wrreliabilitycoordinator@grid-india.in, wrldccr@grid-india.in, venkyminnakuri@grid-india.in**
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Copy to (for information):

1. **SA to Chairperson, CEA**
2. **SA to Member (PS), CEA**
3. **SA to Member (GO&D), CEA**

Power System Engineering & Technology Development Division
Central Electricity Authority

Minutes of the Meeting to deliberate on Tripping of 400 kV Transmission Lines of Mumbai Area on 30-31 July, 2025:

1. An on-line meeting was held under the chairmanship of Member (Power Systems), CEA on 04.08.2025 at 11:30 hrs to deliberate on the tripping of 400 kV Transmission Lines feeding Mumbai Area on 30-31 July, 2025. List of participants is given at **Annexure-I.**

2. At the outset, Member (Power Systems), CEA welcomed all the participants and briefed about the agenda of the meeting. He informed that the matter regarding tripping of the following three transmission Lines feeding power to Mumbai area, occurred on 30.07.2025/31.07.2025 was apprised in the HMOP meeting held on 01.08.2025.

a. 400 kV Phadge -Navi Mumbai S/C line
b. 400 kV Phadge- Kharghar S/C line
c. 400 kV Navi Mumbai-Vikhroli S/C line

He stated that tripping of the above lines is stated to be due to large no. of Insulator failures. As these lines have been commissioned recently, failure of so many Insulators is a matter of concern. As any further tripping of lines feeding to Mumbai may lead to precarious situation of large-scale interruption of Power Supply, efforts shall be made to restore the above lines on priority basis. Failure of the insulator shall be analyzed by the Standing Committee on the Tower failure/ PSE&TD Division, CEA.

3. He requested WRPC, WRLDC and M/s Resonia (MUML-Sterlite), and M/s Adani (KVTPL-Adani) to explain the issues relating to tripping of the above Lines and reasons for failure of large no. of Insulators within short span of time from their commissioning, along with measures taken to restore them.

4. Representative of WRLDC made a presentation, and the following were highlighted:
   a) 400 kV Padghe (PG)-Kharghar S/c tripped on R-N fault at 03:45 hrs of 30.07.025. 400 kV Padghe(PG)-Navi Mumbai S/c tripped on B-N fault at 12:15 hrs of 30.07.2025 and 400 kV Navi Mumbai – Vikhroli S/c on Y-N fault at 03:24 hrs of 31.07.2025. These lines tripped due to failure of insulators. While the ownership of the first two lines is with MUML-Sterlite, that of the third line is with both MUML-Sterlite and KVTPL-Adani. However, the Insulator failures portion belongs to MUML-Sterlite. And all the three lines were commissioned in September, 2024
   b) Mumbai Metropolitan Region (MMR) area is fed mainly from the seven (7) numbers 400 kV lines and 2x315 MVA + 2x500 MVA, 400/220 kV Boisar ICTs. During tripping of above lines in MMR area, the remaining feed was through following elements:
      - 400 kV Padghe (MH)-Kalwa-D/c lines (HTLS)
      - 400 kV Pune (PG)-Kharghar-S/c (Twin Moose)
      - 400 kV Pune (PG)-Vikhroli-S/c (Twin Moose)
      - 2x315 MVA + 2x500 MVA, 400/220 kV Boisar ICTs
   c) Due to tripping of the said 400 kV lines, the loading on the above elements were significantly increased. Further, drawl of MMR from the grid also observed to have exceeded its ATS limit of 1690 MW. To manage the situation, the internal/ embedded generation from with MMR had been maximized (1877 MW) to cater to MMR's load.

d) From the meetings they held with Sterlite on 31.07.2025, 01.08.2025, and 02.08.2025, it was noted that all the failed insulators are of Tag make, and were from manufacturing batch of December, 2022 and July, 2023.

e) As additional precautionary measures, the following have also been done:
   - MSLDC has been instructed to identify areas for manual load shedding, in addition to existing LTS (Load Trimming Scheme), to control transmission line loadings under stressed conditions.
   - MSLDC/ MSETCL has proposed to take load on 2x500 MVA, 400/220 kV Kudus (MH) ICTs to create an alternate supply path from Padghe (PG) → Kudus → Borivali (Mumbai). This is to be implemented through an interim arrangement of 220 kV Kudus–Borivali-2, by cross-jumpering 220 kV Kudus–Ghodbunder and 220 kV Ghodbunder–Borivali lines, effectively bypassing Ghodbunder. Load taken on 2x500 MVA, 400/220 kV ICTs at 02:56hrs & 02:58hrs of 03-08-2025 respectively after implementation of interim arrangement. However, hotspot on 220 kV Kudus-Borivali-2 (interim arrangement) was observed. 400/220 kV Kudus ICTs also hand tripped at 10:58hrs/03-08-2025 for controlling the loading on 220 kV Kudus-Borivali-1.

f) As on 04.08.2025, 400 kV Phadge- Kharghar line was restored (charged) at 10.01 hrs of 04.08.2025. And the remaining two lines were also expected to be to restored/ charged by 04.08.2025.

g) The preliminary report furnished by Grid-India/ WRLDC in this regard is at **Annexure-II**.

5. Representative of MUML-Sterlite informed that tripping of the lines was due to the failure of the insulators. He informed that 45 Nos. insulator had been replaced in the 400 kV Phadge- Kharghar line; 50 Nos. Insulators (30 failed and 20 suspected) had been replaced in the 400 kV Phadge -Navi Mumbai line; and 13 Nos Insulators had been replaced and 8 Nos. insulators to be replaced in the 400 kV Navi Mumbai-Vikhroli line. He also informed that sample of the failed insulator was also sent to ERDA for testing, and waiting the test results.

   He also informed that 400 kV Phadge- Kharghar S/C line had been restored by 10 hrs on 04.08.2025. The 400 kV Phadge -Navi Mumbai S/C line was expected to be restored by 12:00 hrs on 04.08.2025, and 400 kV Navi Mumbai-Vikhroli S/C line was expected to be restored by late night of 04.08.2025.

6. Representative of the M/s Adani confirmed that there were no failure of insulators in their line.

7. In this regard, Chief Engineer (PSE&TD), CEA stated the following:
   - As so many Insulators (Long-rod Polymer type) were observed to have failed within two years of their manufacturing, the concerned Utilities/ Line Owners should take up the issue with their OEMs, and review the Quality Assurance Plan (QAP) with focus on Insulators being followed by them with the OEMs. As the failed Insulators were observed to be of same ('Tag') make from manufacturing batch of December, 2022 and July, 2023, he requested to collect details regarding supplying of the same lot/batch Insulators to other utilities so that corrective action including their

replacement may be taken up. On this, WRPC confirmed that the same had already been initiated by them, and would be getting the requisite details within a week time.

- In this regard, he briefed about the Model Quality Assurance Plan (MQAP) Guidelines issued by CEA in March, 2022, and requested all concerned (Utilities and OEMs) to adhere to them. He also detailed the regulatory requirements w.r.t. Insulators, including the requirement of maintaining a minimum specific creepage distance 31 mm/ kV (as per IS 16683: Part-I: 2018). MUML-Sterlite, KVTPL-Adani, PGCIL confirmed their compliance w.r.t. the same.
- He also stated that, as suggested by Member (PS), CEA, the above Insulator Failure events would be discussed in the forthcoming meeting of Standing Committee of Experts on Tower Failures, and requested MUML-Sterlite furnish details as per the format given at **Annexure-III**.
- He also flagged the issue of several forced outages in MMR and South Gujarat during the period from 30-07-2025 to 02-08-2025, resulting in emergency operating conditions. Some 13 Nos. of lines belonging to GETCO, PGCIL & MSETCL were observed to have got tripped, out of which some events were observed to be due to Insulator failures. He requested concerned Utilities to take preventive O&M of the lines regularly, and ensure their availability during critical times such as above.
- W.r.t. MSLDC/ MSETCL's failed attempt of providing alternate supply path through Padghe (PG) → Kudus → Borivali (Mumbai) from 220 kV side, he emphasized on the need for regular O&M of substation equipment and transmission lines so that timey detection of weak links & hot spots is ensured, and corrective action is taken in time.

The meeting ended with thanks to the Chair.

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Annexure-I

List of Participants

CEA

1. Sh. V.K. Singh, Member (Power Systems)
2. Sh. N.R.L.K. Prasad, Chief Engineer (PSE&TD)
3. Sh. Bhanwar Singh Meena, Director (PSE&TD)
4. Sh. O.P. Suman, Director (GM)
5. Sh. Pankaj Kumar Verma, Deputy Directory (PSE&TD)

NLDC

1. Sh. Raj Kishan, Dy. Manager
2. Sh. Rahul Shukla, Ch. Manager
3. Sh. Vivek Pandey
4. Sh. Priyam Jain
5. Sh. Mohit Gupta
6. Sh. Gaurab Dash
7. Sh. Abhijeet Prakash

WRPC

1. Sh. Deepak Kumar, Member, Secretary

WRLDC

1. Sh. Mahesh M Mehendale, ED
2. Sh. Goodelli Madhukar
3. Sh. M. Venkateswara Rao
4. Sh. Vijay Kodali
5. Sh. Sh. Akhil Gupta
6. Sh. Sunil Patil
7. Sh. Manoj Thakur
8. Sh. Omkar

POWERGRID

1. Sh. Machha Krishnanjaneyulu

TATA POWER

1. Sh. Ashish Bhat
2. Sh. Ashish Shukla
3. Sh. Devanand Pallikuth
4. Sh. Hemant Karadkar
5. Sh. H. B. K. Patnaik
6. Sh. Milind Gole

RESONIA

1. Sh. Mahesh Bhagat
2. Sh. Rajesh Suri

ADANI

1. Sh. Abhishek Kukreja
2. Sh. Shivraj Chouhan

OTHER PARTICIPANTS

1. Representative from SLDC
2. Sh. Prakash
3. Sh. Shailendra Verma
4. Sh. Abdullah Siddique
5. Sh. S. S. Raghuwanshi

Preliminary report on multiple tripping in transmission lines supplying power to Mumbai Metropolitan Region (MMR) on 30th & 31st July 2025

1. **Date & Time of event:** 30-07-2025 and 31-07-2025

2. **Area Affected:** Mumbai Metropolitan Region (MMR)

3. **GD/GI Category:** Near Miss Incident

4. **Generation loss/Load loss:** Nil.

5. **Tripped elements:**

S. No.	Line Name	Tripping Time	Reason	Number of charging attempts taken	Length (km)	Owner	COD
1	**400 kV Padghe(PG)-Navi Mumbai S/c**	Tripped at 03:45 hrs/30-07-2025.	Transmission Line faults due to failure of insulators at multiple locations	03	68	MUML-Sterlite	**16-09-2024**
2	**400 kV Padghe (PG)-Kharghar S/c**	Tripped at 12:15 hrs/30-07-2025.		03	71	MUML-Sterlite	**24-09-2024**
3	**400 kV Navi Mumbai – Vikhroli S/c**	Tripped at 03:24 hrs of 31-07-2025		01	61	KVTPL-Adani and MUML-Sterlite	**27-09-2024**

6. **Details of charging attempts:**

S. No.	Line Name	Number of charging attempts taken (Complete line length)	Section-wise charging details
1	**400 kV Padghe(PG)-Navi Mumbai S/c**	03	**From Padghe (PG) end:** Charging not attempted section-wise, as Padghe (PG) has a GIS bus. All charging attempts were made for the complete line. POWERGRID requested charging only after rectification of broken insulators to safe guard GIS equipment as already 3 attempts have taken for complete line.

			From Navi Mumbai end: Charging not attempted as 400 kV source is not available. 220 kV lines connected through 400/220 kV ICTs are inadequate to feed fault current, if any.
2	**400 kV Padghe (PG)-Kharghar S/c**	03	**From Padghe (PG) end:** After replacing failed insulators at loc. no. 86/0 and 97/0, MUML-Sterlite test-charged the line up to loc. 56/0 at 21:53 hrs on 01-08-2025. However, the line could not sustain. Further inspection/replacement of insulators is ongoing. **From Kharghar end:** After isolating jumper at loc. 56/0, the line was idle charged from Kharghar at 01:20 hrs on 01-08-2025. No insulator failure was observed.
3	**400 kV Navi Mumbai – Vikhroli S/c**	01	**From Navi Mumbai end:** Charging not attempted. No request was received from MUML-Sterlite. **From Vikhroli end:** Charging not attempted No request was received from MUML-Sterlite.

The above mentioned 400 kV lines tripped during charging attempts and are still under outage.

7. Details of fault (if any):

Nature of fault as per PMU:

a. 400 kV Padghe (PG)-Kharghar S/c tripped on R-N fault with fault clearing time of 520 ms.
b. 400 kV Padghe(PG)-Navi Mumbai S/c tripped on B-N fault with fault clearing time of 80 ms.
c. 400 kV Navi Mumbai – Vikhroli S/c on Y-N fault with fault clearing time of 80 ms.

8. Brief description of event:

a. Background:

i) Mumbai Metropolitan Region (MMR) area is fed mainly from the following 400 kV lines / elements:

S. No	Line Name	Conductor Type	Length (km)	Owner	Thermal Loading Limit (MVA)
1	400 kV Padghe (PG)-Kharghar S/c	Quad Moose	71	MUML-Sterlite	2100

2	400 kV Padghe(PG)-Navi Mumbai S/c	Quad Moose	68	MUML-Sterlite	2100
3	400 kV Padghe(MH)-Kalwa-1	HTLS	52	MSETCL	2100
4	400 kV Padghe(MH)-Kalwa-2	HTLS	52	MSETCL	2100
5	400 kV Pune (PG)-Kharghar S/c	Twin Moose	93	MSETCL/POWERGRID	874
6	400 kV Pune(PG)-Vikhroli S/c	Twin Moose	110	KVTPL – Adani/MSETCL/POWERGRID	874
7	400 kV Navi Mumbai – Vikhroli S/c	Quad Moose	61	KVTPL-Adani and MUML-Sterlite	2100
8	2x315 MVA + 2x500 MVA, 400/220 kV Boisar ICTs	--	-	POWERGRID	

b. Network Connectivity

The power map of showing the connectivity of MMR is given below:

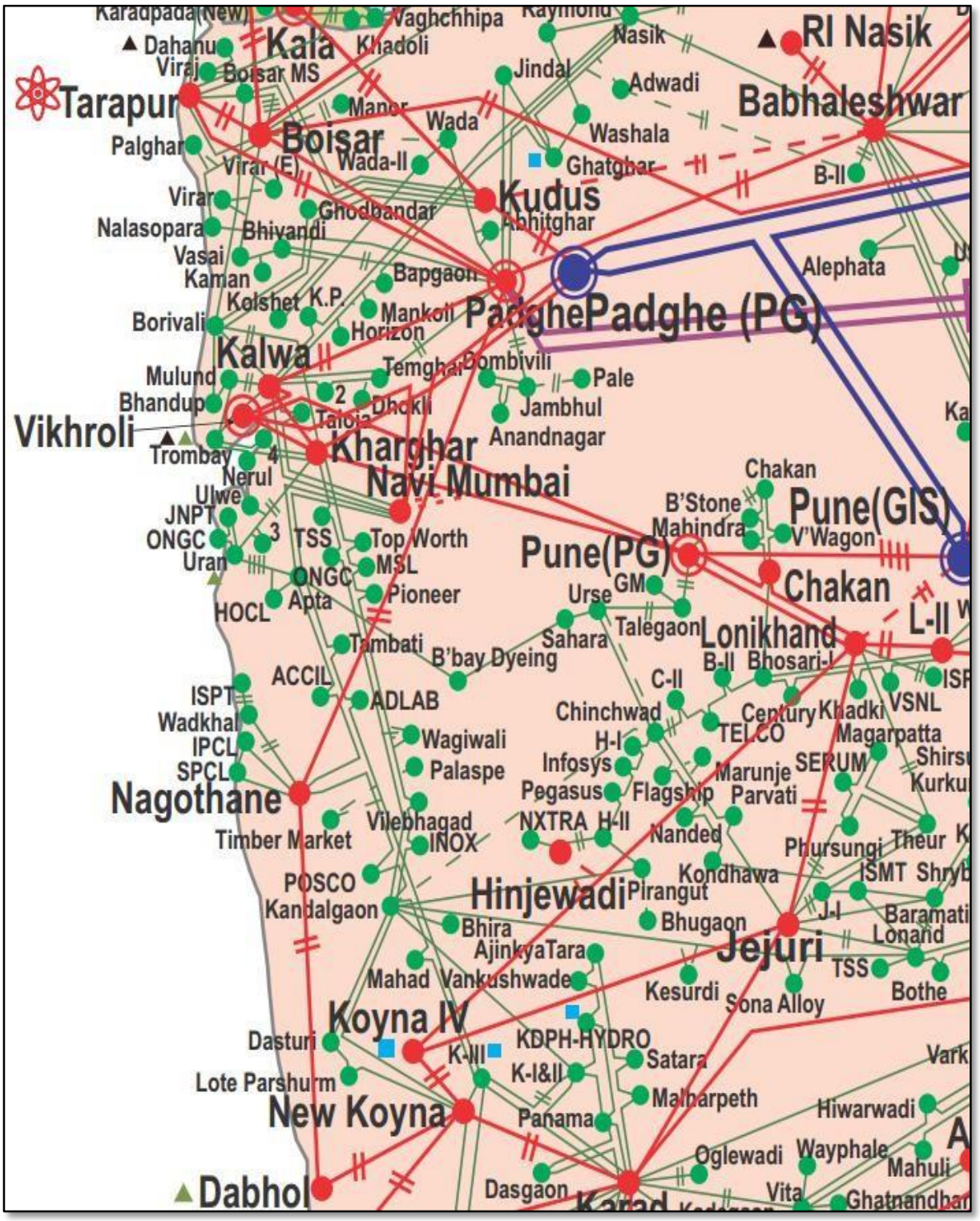


Figure 1: Power Map of MMR area

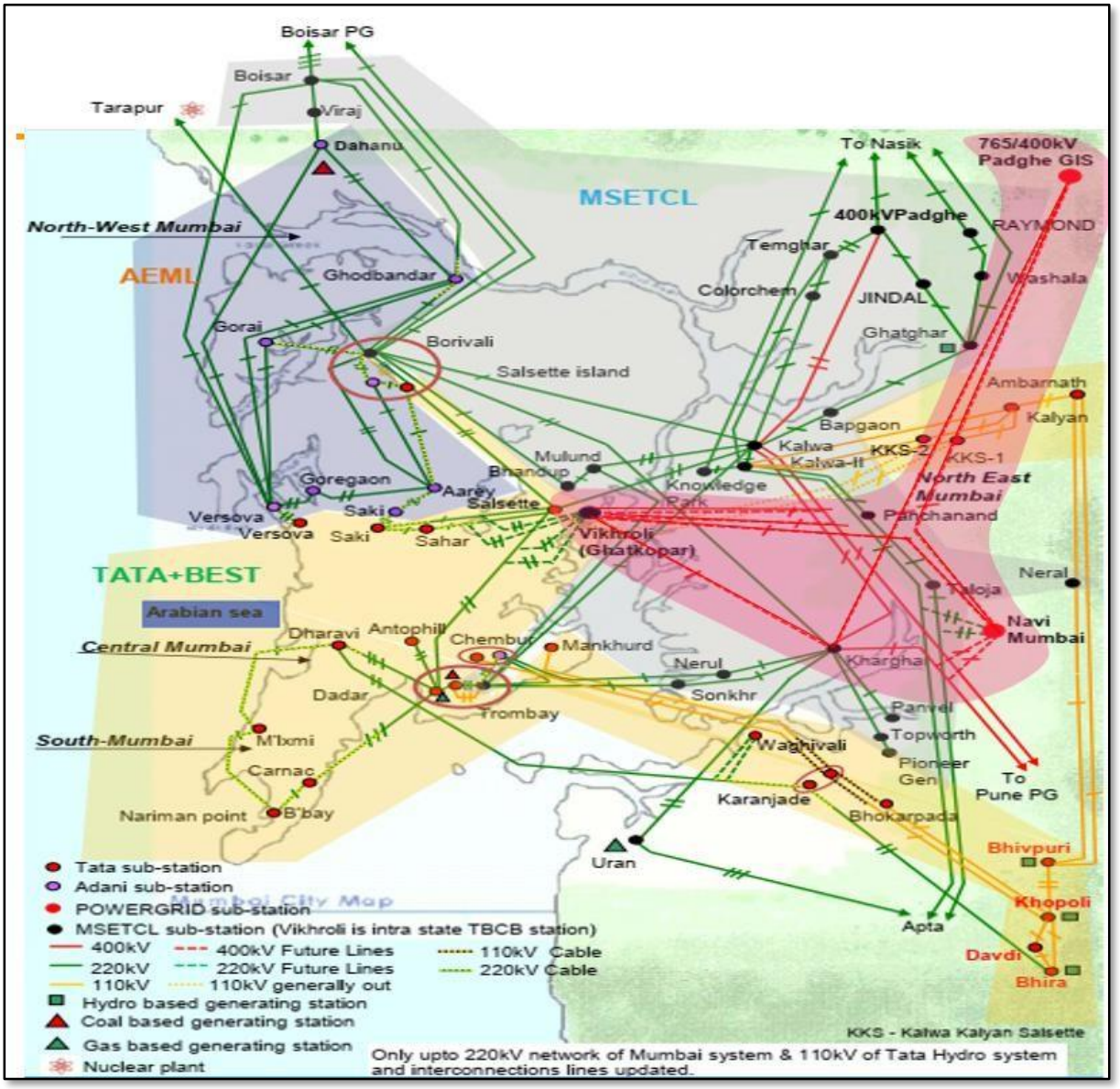


Figure 2: Power Map showing sub-systems of Mumbai power network

c. Antecedent Conditions

Western Region was meeting a demand of 54584 MW prior to the event. The summary of demand and generation in Maharashtra & MMR is given below in Table 1 – 3:

Table 1 Demand vs Generation

Control Area	Demand (MW)	Internal Generation (MW)	Drawl (MW)
Maharashtra	21459	13329	8130
Mumbai Metropolitan Region	3392	964	2428

Table 2 Generation availability in MMR area at 12:10Hrs/30-07-2025

Sr. No.	Station name	Type	Installed Capacity (MW)	On-bar Capacity (MW)	Actual Generation (MW)	Actual MVAr
1	Trombay Thermal (Tata)	Thermal (Intra-state)	930	930	564	Not available
2	Dahanu (AEML)		500	250	250	
3	Tata Hydro	Hydro (Intra-state)	447	222	150	32
4	Uran	Gas (Intra-state)	672	228	121	-29
5	Ghatghar	Hydro (Intra-state)	250	0	0	0
6	TAPS(1&2)	Nuclear (ISGS)	320	0	0	0
7	TAPS(3&4)	Nuclear (ISGS)	1080	1080	975	80
	Total			**2710**	**2060**	**83**

Table 3 Generation under outage in MMR area on 30-07-2025

Sr. No.	Station name	Installed Capacity (MW)	Type of Outage	Reason
Inter State Generating Station (ISGS)				
1	TAPS 1&2 (Nuclear) - Unit-1&2 (2x160 MW)	320	Planned	Unit out since 08-Jan-2020 for repair works and refuelling, Unit-2 out since 13-July-2020 (Initially was out for cleaning of clogged basket strainer of cooling water system due to heavy ingress of debris from sea/ The outage extended for primary system piping replacement and refuelling from 17-Mar-2022).
Total		320		

Sr. No.	Station name	Installed Capacity (MW)	Type of Outage	Reason
Intra-State Generation				
1	Dahanu Unit-1	250	Planned	AOH
Total		250		

d. Sequence of Events (SOE):

Table 4 Details of SOE

Date	Time	Element Name	Reason
30-07-2025	03:45 hr	400 kV Padghe (PG) – Navi Mumbai S/c line tripped	B-E fault
	07:45 hr	Charging attempt failed in 400 kV Padghe (PG) – Navi Mumbai S/c	SOTF
	12:15 hr	400 kV Padghe (PG) – Kharghar S/c line tripped with 400 kV Bus-1 at Padghe (PG)	R-E fault
	14:18 hr	400 kV Bus-1 at Padghe (PG) charged	
	14:52 hr	Charging attempt failed in 400 kV Padghe (PG) – Kharghar S/c	SOTF
	16:41 hr	400 kV Padghe (PG)-Navi Mumbai S/c line charged after insulator replacement.	
	16:48 hr	400 kV Padghe (PG) -Navi Mumbai S/c line again tripped	Y-E fault
31-07-2025	03:07 hr	400 kV Padghe (PG) -Kharghar S/c line charged.	
	03:10 hr	400 kV Padghe (PG) -Kharghar S/c line tripped	R-E fault
	03:24 hr	400 kV Navi Mumbai – Vikhroli S/c line	Y-E fault

Date	Time	Element Name	Reason
		tripped	
	06:40 hr	Charging attempt failed in 400 kV Navi Mumbai – Vikhroli S/c	SOTF
	08:04 hr	Charging attempt failed in 400 kV Padghe (PG) -Navi Mumbai S/c.	SOTF
	13:07 hr	400 kV Padghe (PG)-Kharghar S/c line charged again.	
	13:09 hr	400 kV Padghe (PG)-Kharghar S/c line tripped again	B-E fault

The details about above sequence is enclosed as Annex-I. The PMU plots indicating the fault on the transmission lines is enclosed as Annex-II.

e. Demand Trend of Mumbai Area

The peak demand of Mumbai region over the past 10 days has been around 3500 MW. During this period, the maximum internal generation was ~1500 MW and maximum drawl from the grid was ~2750 MW. It could be observed that MMR drawl exceeded its ATC limit of 1690 MW during morning hours on 30th & 31st July 2025. Embedded generation capacity in Mumbai system is 1877 MW (approx. 50% of load). Dahanu Unit #1 (250 W) is under planned outage since 13-07-2025 & expected to revive by 22-08-2025. The demand & generation pattern plot is given below in Figure 3.

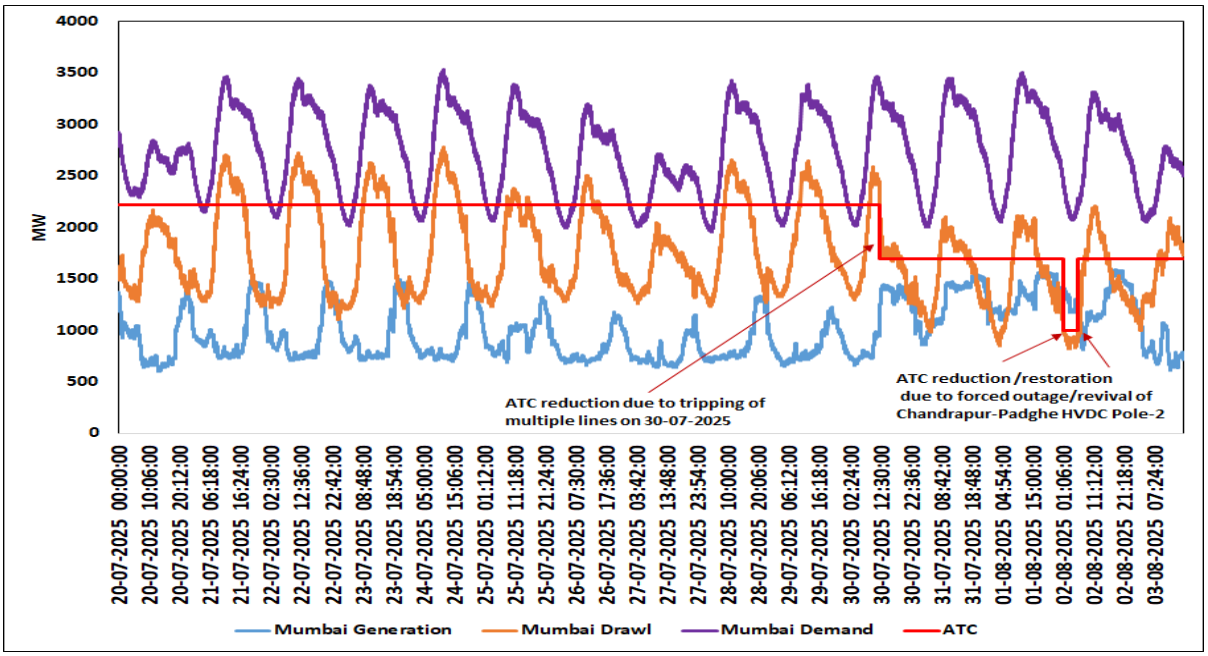


Figure 3: MMR Demand and Generation

f. Constraints in import of Mumbai Area

With the above forced outages in MMR area, the remaining feed is through following elements:

a) 400 kV Padghe (MH)-Kalwa-D/c lines (HTLS)
b) 400 kV Pune (PG)-Kharghar-S/c (Twin Moose)
c) 400 kV Pune (PG)-Vikhroli-S/c (Twin Moose)
d) 2x315 MVA + 2x500 MVA, 400/220 kV Boisar ICTs

The loadings on these elements have increased due to non-availability of lines. The time series trend of loading on these lines as gathered from SCADA is given at Annex-III.

9. Important forced/Emergency outages in MMR and South Gujarat area (30-07-2025-02-08-2025):

In addition to the above forced outages, several forced outages occurred in MMR and South Gujarat during the period from 30-07-2025 to 02-08-2025, resulting in emergency operating conditions. The details of key forced/emergency outages are summarized in table 5.

Table 5 Important forced outages in MMR and South Gujarat area from 30-08-2025 to 02-08-2025

S No	Element Name	Owner	Outage Time	Revival Time	Reason
1	400 kV Vav-Navsari(PG)-S/c	GETCO	08:30Hrs/ 30-07-2025	19:52Hrs/ 30-07-2025	A/R to Non-Auto for attending fault of 400KV Vav-Gandhar B-ph suspension SRI flash at location no- 8 (S) & B-ph tension SRI broken at location no-19(T).
2	Main bay (407) of 400 kV Padghe (PG)-Kharghar-S/c at Padghe (PG) end	POWERGRID-WR1 (PGCIL)	12:15 Hrs/30-07-2025	Still out	400 kV Bus-1 at Padghe (PG) along with 400 kV Padghe (PG)-Kharghar S/c (main and tie bay) tripped at 12:15 hrs/30-07-2025 on R-E fault. CB pole replacement in Rph of 407 Bay is under process and bay will be normalized by 20:00hrs of 04-08-2025.
3	400 kV Gandhar (Jhanor)-Vav-S/c	GETCO	19:50Hrs/ 30-07-2025	19:00Hrs/ 31-07-2025	Line was under forced outage from 01:36Hrs/28-07-2025. Test trial taken from Vav end at 19:50 hrs and line again tripped. Vav: Y-E,Z1,24.434KA,1.593KM.

4	400 kV Aurangabad-M (Waluj)-Pune(GIS)-1	POWERGRID-WR1 (PGCIL), WTPL-Adani, MSETCL	16:28Hrs/ 31-07-2025	19:53Hrs/ 31-07-2025	Emergency outage required on 400 kV Waluj-Pune GIS Ckt-1 at Waluj end to attend hotspot at Y-phase 89B isolator Hanger L nut bolt. Line kept open from 19:53Hrs/31-07-2025 to control the loading on 400 kV Pune(PG)-Chakan-S/c
5	400 kV Ukai-Vav-S/c	GETCO	09:51Hrs/ 01-08-2025	16:03Hrs/ 01-08-2025	To replace Y-Ph broken SRI insulator at Loc No. 63(T).
6	400 kV Vav-Kosamba-S/c	GETCO	10:15Hrs/ 01-08-2025	17:10Hrs/ 01-08-2025	A/R to Non-auto for replacing Y-Ph broken SRI insulator at Loc No. 63(T) on 400 kV Ukai-Vav-S/c
7	400 kV Gandhar (Jhanor)-Vav-S/c	GETCO	15:16Hrs/ 01-08-2025	21:55Hrs/ 01-08-2025	B-phase isolator melting was observed at Gandhar end and on emergency line was opened by Gandhar-NTPC at 15:16 hours at Gandhar end.
8	400 kV Pune (PG)-Chakan-S/c	MSETCL, WTPL-Adani, POWERGRID	00:14Hrs/ 02-08-2025	03:59Hrs/ 02-08-2025	To attend the Hotspot at Line Isolator 'Y' phase of 400 kV Pune (PG)-Chakan Line (Hotspot is 136 Degree C) at Pune (PG) Substation.
9	+/- 500 kV Chandrapur-Padghe HVDC Pole-2	MSETCL	00:47Hrs/ 02-08-2025	05:58Hrs/ 02-08-2025	The HVDC pole 2 got tripped on 02.08.2025 at 00.47hrs due to DC differential current trip Z2. At the time of tripping there was heavy rain fall in DC Yard. Flashover and tracking observed in pole 2 dc yard.

10	400 kV Padghe (PG) Bus-1	POWERGRID-WR1 (PGCIL)	08:41Hrs/ 02-08-2025	02:19Hrs/ 03-08-2025	Emergency Outage of400KV Bus#1 & 400KV Padghe(PG)-Kharghar line (408 and 407Bays to be kept open) is required for CB pole replacement in Rph of 407 Bay (Main bay of 400 kV Padghe (PG)-Kharghar-S/c). Outage of 407 bay will be returned on 20:00hrs of 04-08-2025.
11	400 kV Pune(GIS)-PUNE(PG)-1	POWERGRID-WR1 (PGCIL), WTPL-Adani	16:53Hrs/ 02-08-2025	23:35Hrs/ 02-08-2025	Tripped due to Y-E fault
12	400 kV Pune (PG)-Vikhroli-S/c	KVTPL, MSETCL POWERGRID	16:53Hrs/ 02-08-2025	17:20Hrs/ 02-08-2025	Tripped at Vikhroli end only on Y-E fault, Fault current: 2.66 kA. Line tripped during the fault in 400 kV Pune(GIS)- Pune(PG)-1
13	400 kV Pune(GIS)-PUNE(PG)-2	POWERGRID-WR1 (PGCIL), WTPL-Adani	19:49Hrs/ 02-08-2025	23:22Hrs/ 02-08-2025	A/R Non Auto / A/R disable code to attend the broken insulator of ckt 1

10. Actions taken :

a. During meeting with Sterlite team on 31-07-2025, 01-08-2025 and 02-08-2025, Sterlite informed that the rectification of damaged insulators is being done.
b. Sterlite also informed that failed insulators were observed to be of manufacturing batch (Long rod polymer insulator of Tag make) of Dec'22 and July'23.
c. Due to forced outages, the available transfer capability of Mumbai area has been reduced suitably (1690 MW from earlier 2220 MW) in view of the above forced outages.
d. Embedded generation within Mumbai is being maintained at maximum available capacity.
e. Generation from Uran station is being maintained in the range of 270-360 MW.
f. Ghatghar hydro generation is being brought on bar based on real-time system requirements
g. All generating stations in the MMR area have been instructed to regulate reactive power exchange to aid in voltage control across the system.
h. No planned shutdowns are currently being facilitated in the MMR area

i. MSLDC has been instructed to identify areas for manual load shedding, in addition to existing LTS (Load Trimming Scheme), to control transmission line loadings under stressed conditions.

j. MSLDC/MSETCL has proposed to take load on 2x500 MVA, 400/220 kV Kudus (MH) ICTs to create an alternate supply path from Padghe (PG) → Kudus → Borivali (Mumbai). This is to be implemented through an interim arrangement of 220 kV Kudus-Borivali-2, by cross-jumpering 220 kV Kudus-Ghodbunder and 220 kV Ghodbunder-Borivali lines, effectively bypassing Ghodbunder. Load taken on 2x500 MVA, 400/220 kV ICTs at 02:56hrs & 02:58hrs of 03-08-2025 respectively after implementation of interim arrangement. For attending the hotspot on 220 kV Kudus-Borivali-2 (interim arrangement), 400/220 kV Kudus ICTs also hand tripped at 10:58hrs/03-08-2025 for controlling the loading on 220 kV Kudus-Borivali-1.

Annex-I: Details of Sequence of Events

30-07-2025:

i) 400 kV Padghe (PG)-Navi Mumbai S/c tripped at 03:45 hrs/30-07-2025 on B-E fault. Charging readiness given by MUM-Sterlite at 07:22 hrs/30-07-2025 stating that no abnormality found and line ready for charging. Code issued and line charging attempted at 07:45 hrs, line did not hold.

ii) At 11:30 hrs, Sterlite informed that during emergency patrolling, they have found 1 nos. punctured insulators of B-Phase at Location no. 70/1 and requested work permit. Work permit code issued to MUM-Sterlite for 400 kV Padghe (PG)- Navi Mumbai S/c at around 11:32 hrs.

iii) 400 kV Padghe (PG)-Kharghar S/c tripped (along with 400 kV Bus-1 at Padghe PG) at 12:15 hrs/30-07-2025 on R-E fault. 400 kV Bus-1 charged at 14:18 hrs. Line charging attempted at 14:52 hrs but line tripped on SOTF.

iv) At 18:16 hrs, mail received from MUM-Sterlite stating that during emergency patrolling, found 4 nos. punctured insulators of R-Phase at Location no. 84A/2. Work permit issued at 18:18 hrs.

v) 400 kV Padghe (PG)-Navi Mumbai S/c charging readiness given at 15:56 hrs after attending punctured insulator by MUM-Sterlite. Line charged at 16:41 hrs. Line again tripped at 16:48 hrs on Y-E fault.

vi) At 20:55 hrs, Sterlite requested work permit to do emergency Tower top patrolling at location no. 90/0 to 90A/0 from 21:00 Hrs to 24:00 Hrs. Work permit Code issued at 20:55 hrs.

31-07-2025

vii) 400 kV Padghe (PG)-Kharghar S/c charging readiness received at 02:12 hrs/31-07-2025 after Insulator Replacement work at Loc. No. 84A/2. Line charged at 03:07 hrs. But line again tripped at 03:10 hrs on R-E fault.

viii) 400 kV Navi Mumbai-Vikhroli S/c tripped at 03:24 hrs on Y-E fault. At 05:45 hrs, mail received from Adani that Line Patrolling done by KVTPL-Adani & Sterlite-MUML line team, found no abnormality & requested to take a trial attempt. At 06:15 hrs, test charging code issued. At ~06:40 hrs, line test charging attempted but did not hold.

ix) 400 kV Padghe (PG)-Navi Mumbai S/c charging readiness received at 07:12 hrs after completion of emergency Tower top patrolling & Insulator Replacement work at Location No. 94/0. Line charging attempted at 08:04 hrs but line did not hold.

x) 400 kV Padghe (PG)-Kharghar S/c charging readiness received at 11:49 hrs after Insulator Replacement work at **Loc. No. 78/0**. Code issued and line charged at 13:07 hrs. But line again tripped at 13:09 hrs.

01-08-2025

xi) MUML-Sterlite informed that the defective insulators are observed in the section from Padghe (PG) to tower location 56/0 for both 400 kV Padghe (PG)-Khargar S/c and 400 kV Padghe (PG)-Navi Mumbai S/c lines (~37 km). After opening jumper at Loc No. 56/0, 400 kV Padghe(PG)- Kharghar S/c was idle charged from Kharghar end at 01:20 hrs/01-08-2025 and no failure of insulators was observed. (From Padghe (PG) to location 56/0, both 400 kV Padghe (PG)-Kharghar S/c and 400 kV Padghe(PG)-Navi Mumbai(PG) S/c are on same tower).

xii) After replacement of failed insulators at loc no. 86/0 and 97/0, MUML-Sterlite had test charged 400 kV Padghe(PG)-Kharghar S/c from Padghe (PG) to loc 56/0 at 21:53 hrs/01-08-2025. However, line could not hold. Further inspection of insulators/replacement is in progress.

02-08-2025

xiii) MUML-Sterlite had informed in the meeting held on 02-08-25 that now the inspection of the faulty section is being taken up again for identification of faulty insulators in 400 kV Padghe(PG)-Kharghar-S/c and 400 kV Padghe(PG)-Navi Mumbai S/c. Inspection of the line and replacement of faulty insulators is going on in parallel and line test charging will be requested after completion of this work.

xiv) Two teams (4 fitters each) are also deployed for identification of faulty insulators in the 400 kV Navi Mumbai-Vikhroli S/c line but priority is restoration of 400 kV Padghe (PG)-Kharghar S/c and 400 kV Padghe(PG)-Navi Mumbai S/c

xv) MUML-Sterlite informed that monkey patrolling of 56 out of 70 locations have been completed and remaining is expected to be completed by 03-08-2025.

03-08-2025

xvi) **400 kV Padghe (PG)-Kharghar-S/c:** MUML-Sterlite informed in the meeting held on 03-08-2025 that patrolling of 69 out of 70 tension tower locations was completed. 44 insulators found to be faulty and 36 insulators replacement is done.

xvii) Patrolling of 38 suspension towers is also being carried out and expected to complete by 03-08-2025. After completion of the work, 400 kV Padghe (PG)-Khargar S/c will be test charged from 400 kV Padghe (PG) to location No. 56/0 to check the healthiness. If found healthy, line is expected to be charged by 03-08-2025

xviii) **400 kV Padghe (PG)-Navi Mumbai-S/c:** 400 kV Padghe (PG)-Khargar S/c and 400 kV Padghe (PG)-Navi Munbai S/c are double ckt till tower 56/0, monkey patrolling of 400 kV Padghe (PG)-Navi mumbai S/c upto 56/0 from Padghe (PG) end is also done for 69 out of 70 tension tower locations. 8 insulators were

found faulty so far and all are replaced. Remaining portion i.e. from tower 56/0 to Navi Mumbai, patrolling will be completed by 04-08-2025 and line is expected to be charged on 04-08-2025.

xix) **400 kV Navi Mumbai-Vikhroli-S/c:** Patrolling of 15 tension tower locations out of 127 are completed (MUML portion) and remaining portion is under process. KVTPL-Adani informed that the patrolling of their section is done and no insulators were found faulty. Line is expected to be charged on 05-08-2025.

Annexe-II: PMU based plots indicating the fault in Transmission lines

PMU Data Analysis

30-07-2025

i) **03:45 hr:** 400 kV Padghe (PG) – Navi Mumbai S/c line tripped due to B-Phase fault with an 80 ms clearing time. Auto reclosure attempt was unsuccessful in B-phase. PMU plot is given below:

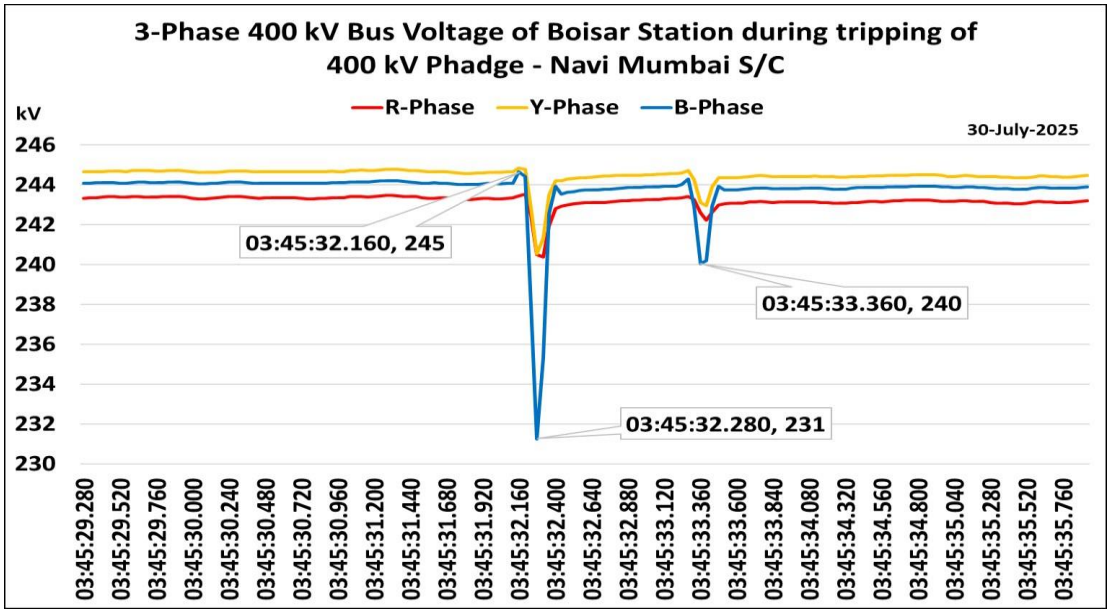


Figure 4: 3-Phase 400 kV Bus Voltage of Boisar Station during tripping of 400 kV Phadge (PG) - Navi Mumbai S/c

ii) **07:45 hr:** 400 kV Padghe (PG) – Navi Mumbai S/c tripped on B-phase again during charging attempt. PMU plot is given below:

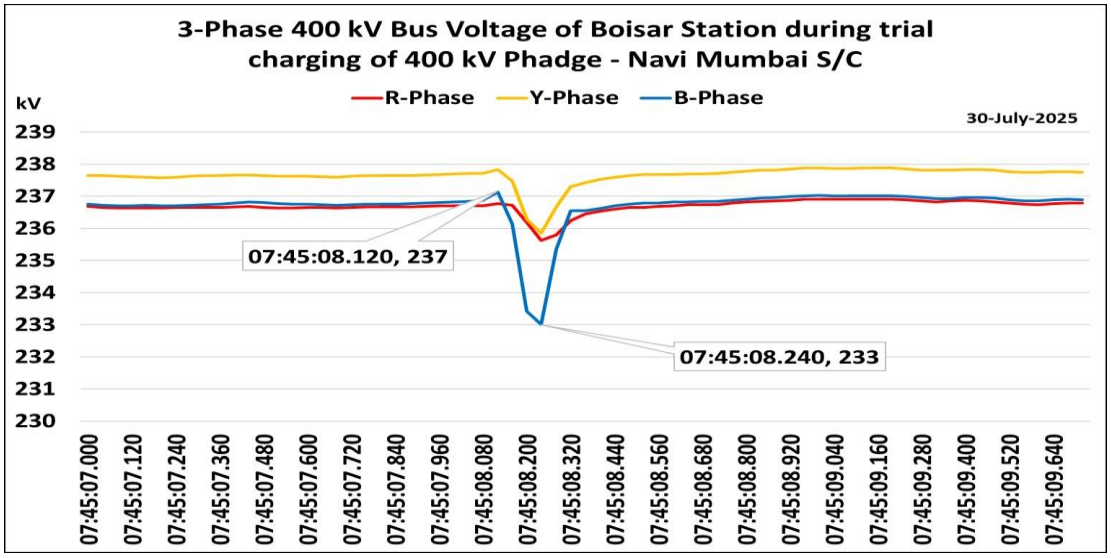


Figure 5: 3-Phase 400 kV Bus Voltage of Boisar Station during trial charging of 400 kV Phadge (PG) - Navi Mumbai S/c

iii) **12:15 hr:** 400 kV Padghe (PG) – Kharghar S/c line tripped along with 400 kV Bus-2 at Padghe PG on R phase fault . The fault clearing time was 520 ms. PMU plot is given below:

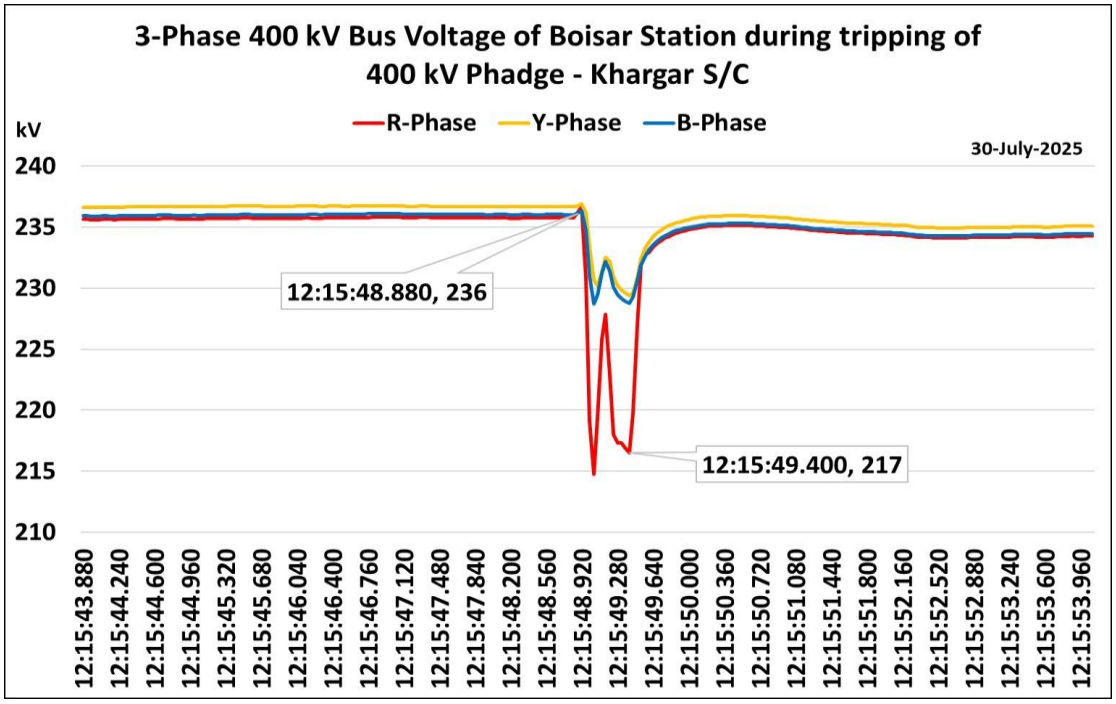


Figure 6:3-Phase 400 kV Bus Voltage of Boisar Station during tripping of 400 kV Phadge (PG) - Khargar S/c

iv) **14:52 hr:** 400 kV Padghe (PG) – Kharghar S/c tripped on R-phase again during charging attempt. PMU plot is given below.

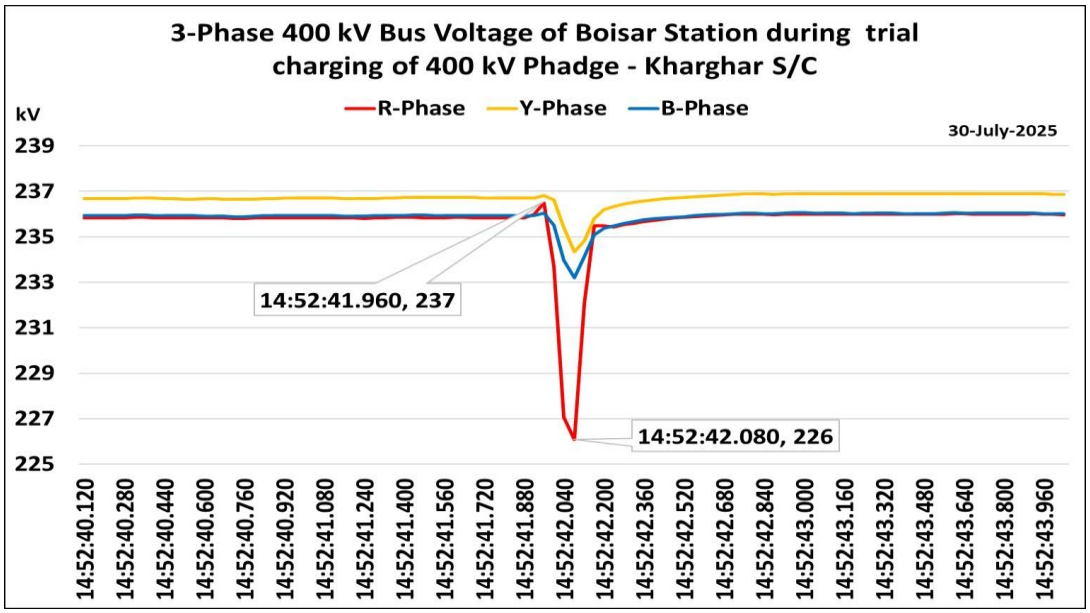


Figure 7: 3-Phase 400 kV Bus Voltage of Boisar Station during trial charging of 400 kV Phadge (PG) - Kharghar S/c

v) **16:48 hr:** 400 kV Padghe (PG)-Navi Mumbai S/c line tripped on Y – Phase fault with fault clearing time of almost 120 ms. Auto reclosure attempt was unsuccessful in Y-phase. PMU plot is given below.

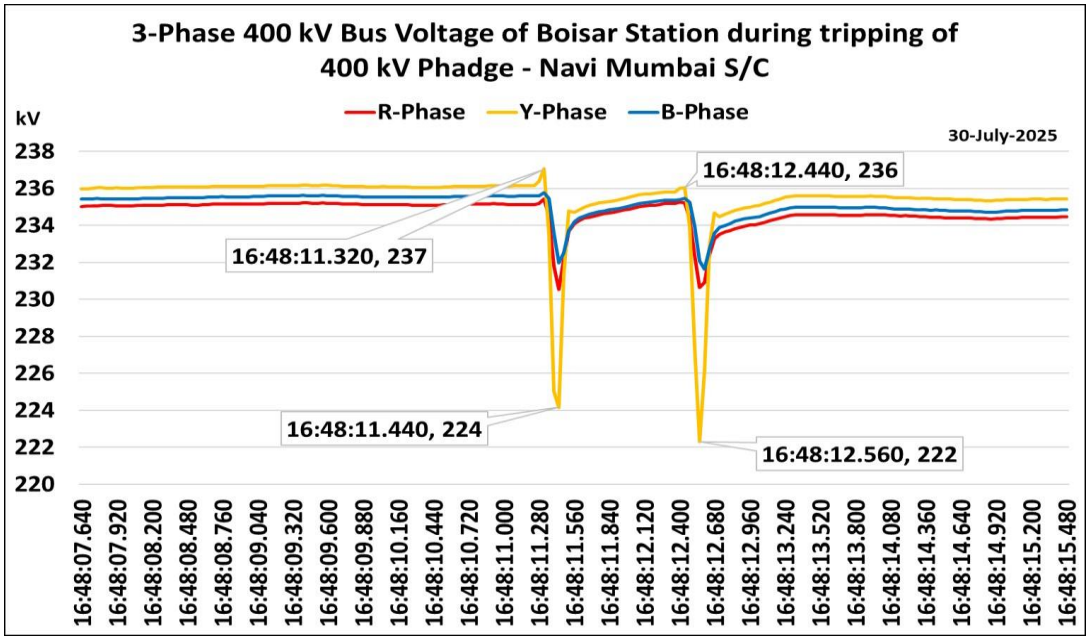


Figure 8: 3-Phase 400 kV Bus Voltage of Boisar Station during tripping of 400 kV Phadge (PG) - Navi Mumbai S/c

31-07-2025

vi) **03:10 hr:** 400 kV Padghe-Kharghar line tripped due to R-Phase fault with an 120 ms clearing time. Auto reclosure attempt was unsuccessful in R-phase. PMU plot is given below:

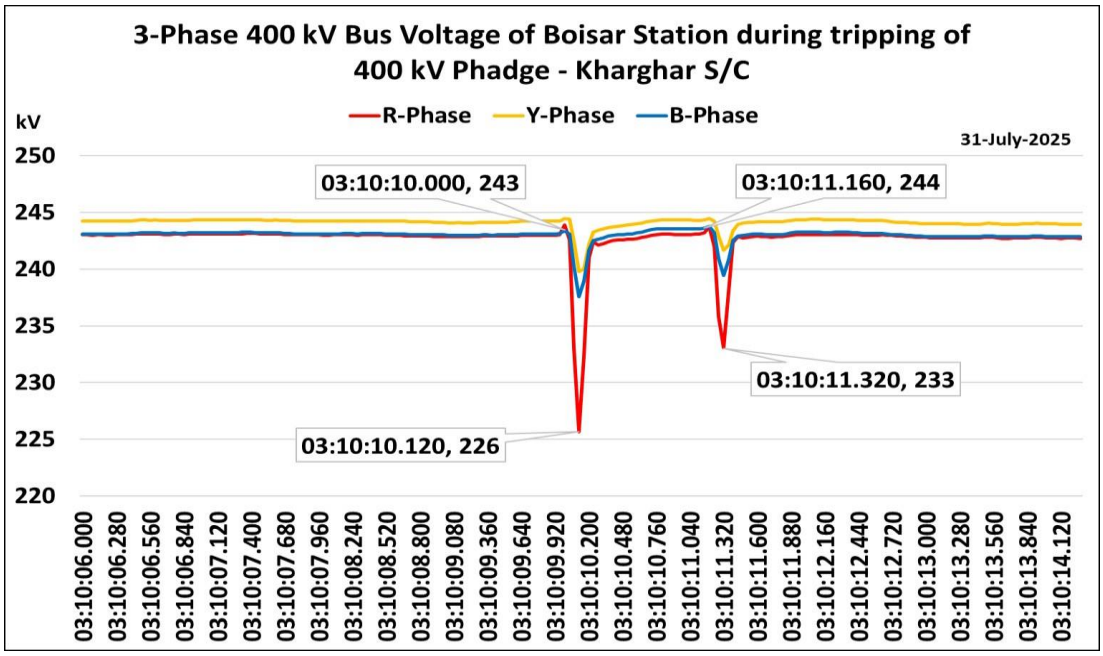


Figure 9: 3-Phase 400 kV Bus Voltage of Boisar Station during tripping of 400 kV Phadge (PG) - Kharghar S/c

vii) **03:24 hr:** 400 kV Navi Mumbai – Vikhroli S/c line tripped on Y-phase fault. Line has not taken A/R attempt in single phase fault.

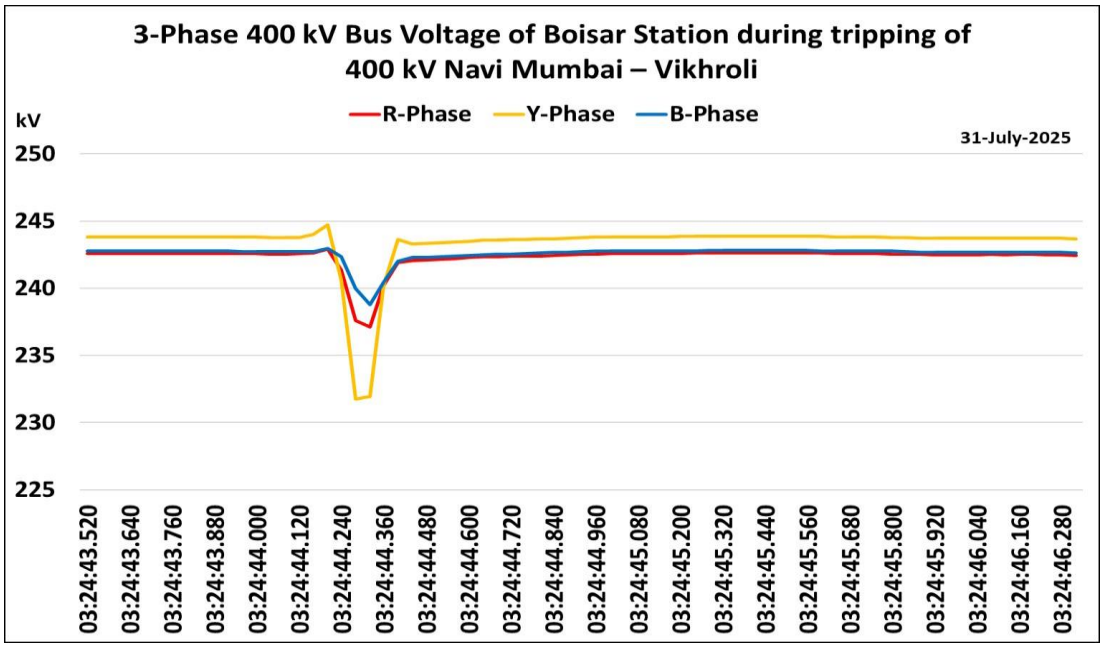


Figure 10: 3-Phase 400 kV Bus Voltage of Boisar Station during tripping of 400 kV Navi Mumbai – Vikhroli

viii) **06:40 hr:** 400 kV Navi Mumbai – Vikhroli tripped during trial charging attempt. As per PMU, no major fault signature was captured during the same time.

ix) **08:04 hr:** 400 kV Padghe (PG)-Navi Mumbai S/c tripped during trial charging attempt. As per PMU fault clearing time was 560 ms.

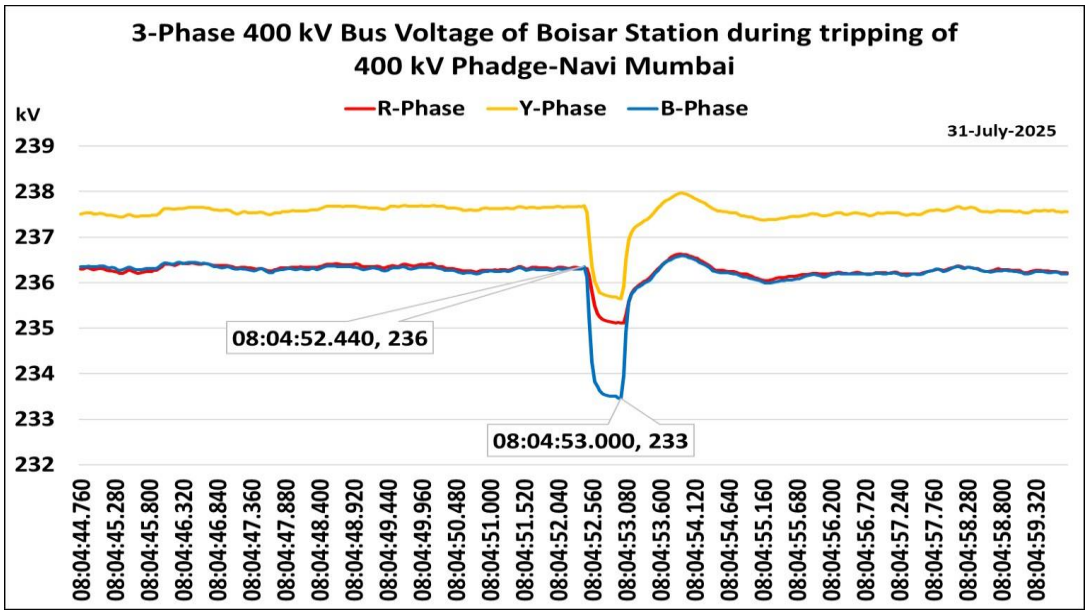


Figure 11: 3-Phase 400 kV Bus Voltage of Boisar Station during tripping of 400 kV Phadge (PG)-Navi Mumbai S/c

x) **13:09 hr:** 400 kV Padghe-Kharghar line tripped due to B-Phase fault with an 80 ms clearing time. Auto reclosure attempt was unsuccessful in B-phase. PMU plot is given below

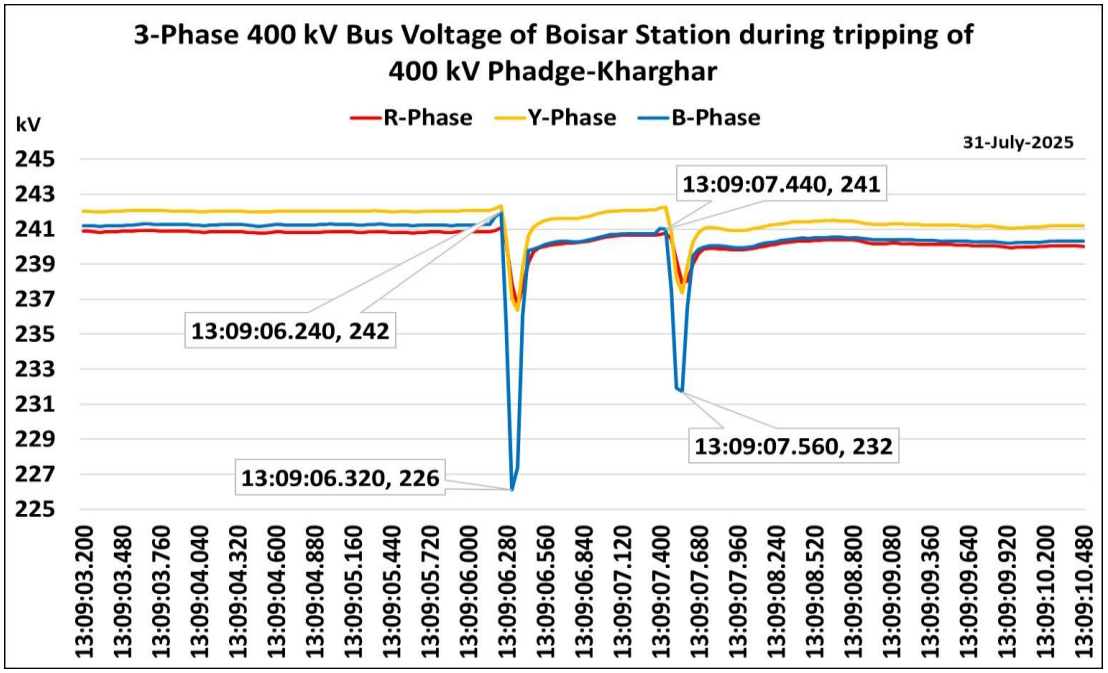


Figure 12: 3-Phase 400 kV Bus Voltage of Boisar Station during tripping of 400 kV Phadge (PG)-Kharghar S/c

Annex-III: The loadings on transmission lines supplying power to MMR as obtained from SCADA Data

i) The power flow on 400 kV lines from Vikhroli station is given below

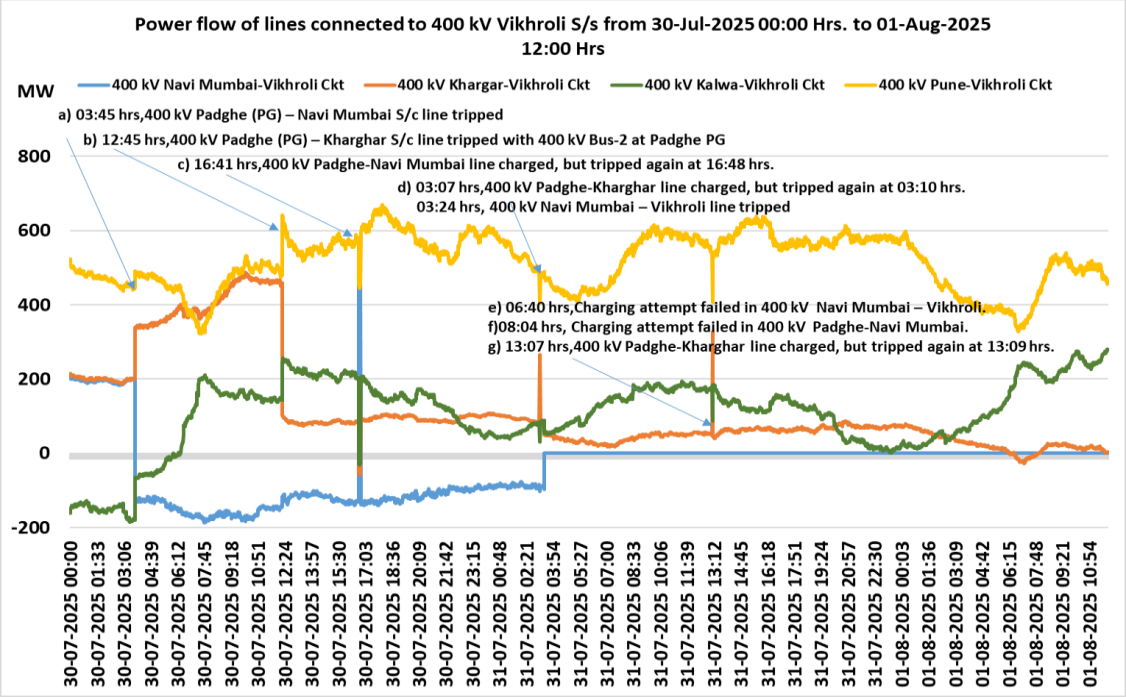


Figure 13: Vikhroli Station SCADA Data

ii) N-1 non-compliance of 400 kV Pune (PG)-Kharghar S/c and 400 kV Pune (PG)-Vikhroli S/c was observed

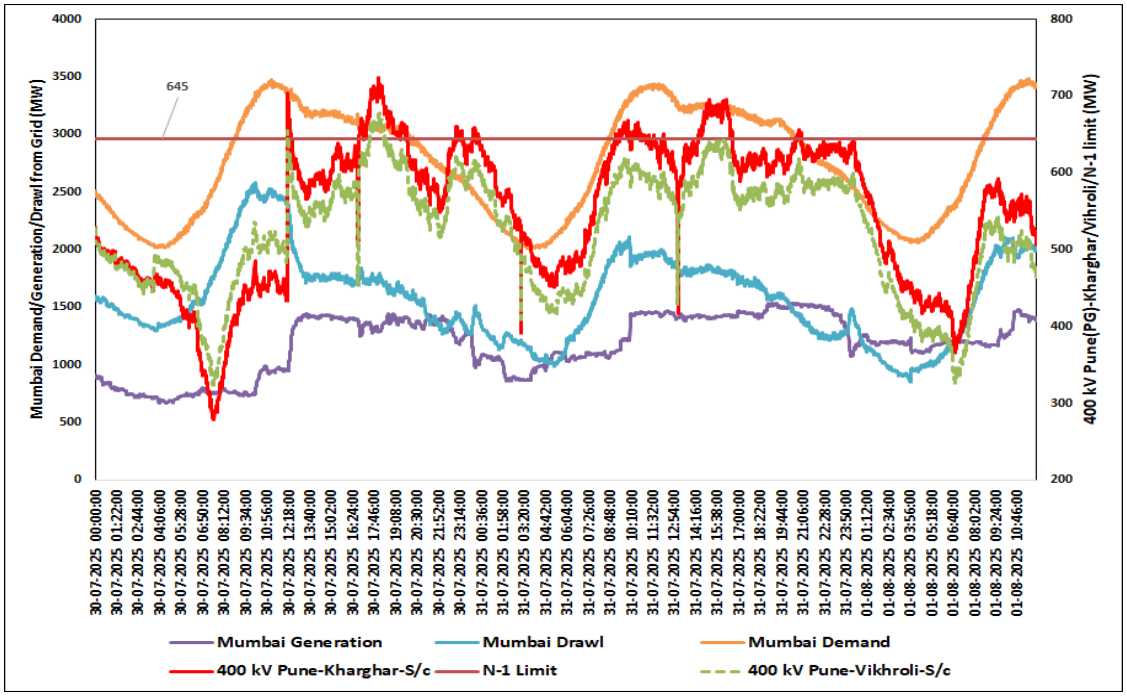
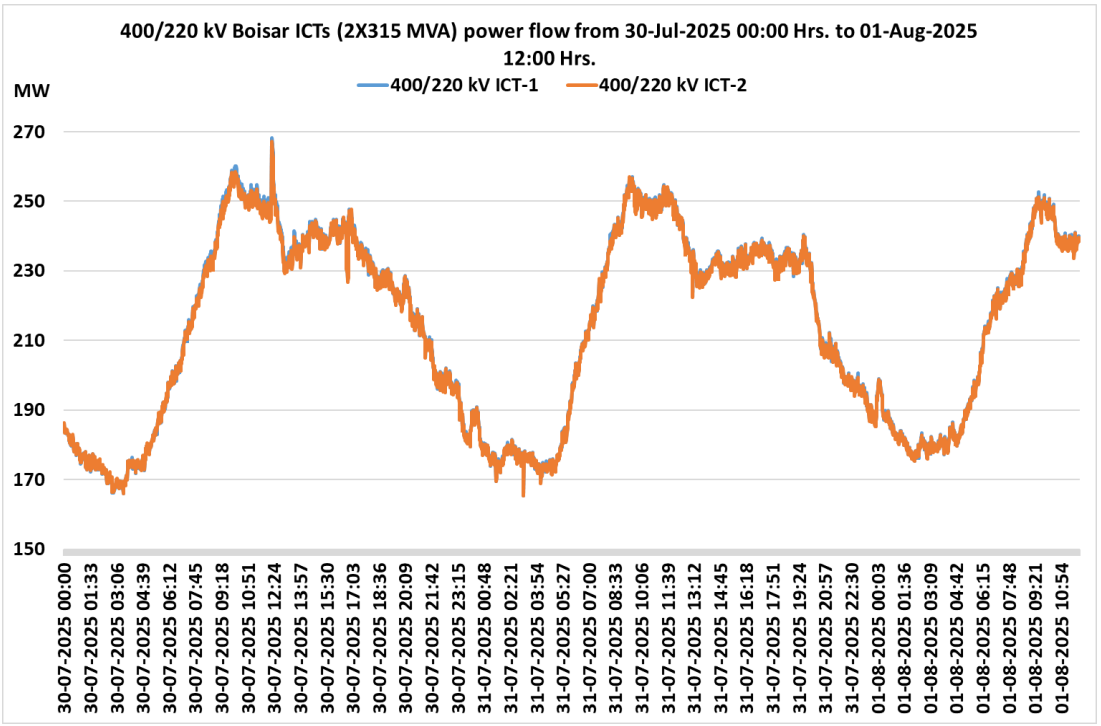


Figure 14: Mumbai Demand/Generation/Drawl vs 400 kV Pune (PG)-Kharghar/Vikhroli lines loading

iii) Power flow on ICT's at Boisar and Phadge station is given below. It could be observed that 400/220 kV ICTs 1 & 2 (315 MVA each) at Boisar station were operating at around 250 MW each, while ICTs 3 & 4 (500 MVA each) were loaded in the range of approximately 400 MW during the period from 09:00 hrs to 12:00 hrs. These ICTs appear to be operating in a non-N-1 compliant manner under these conditions.



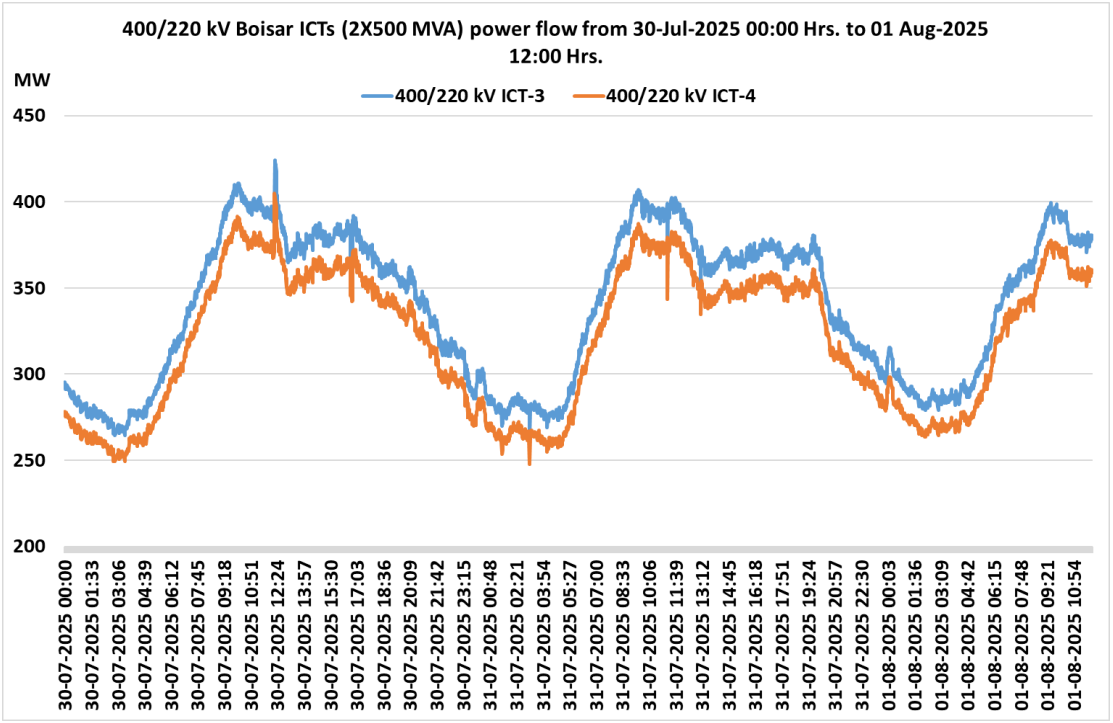
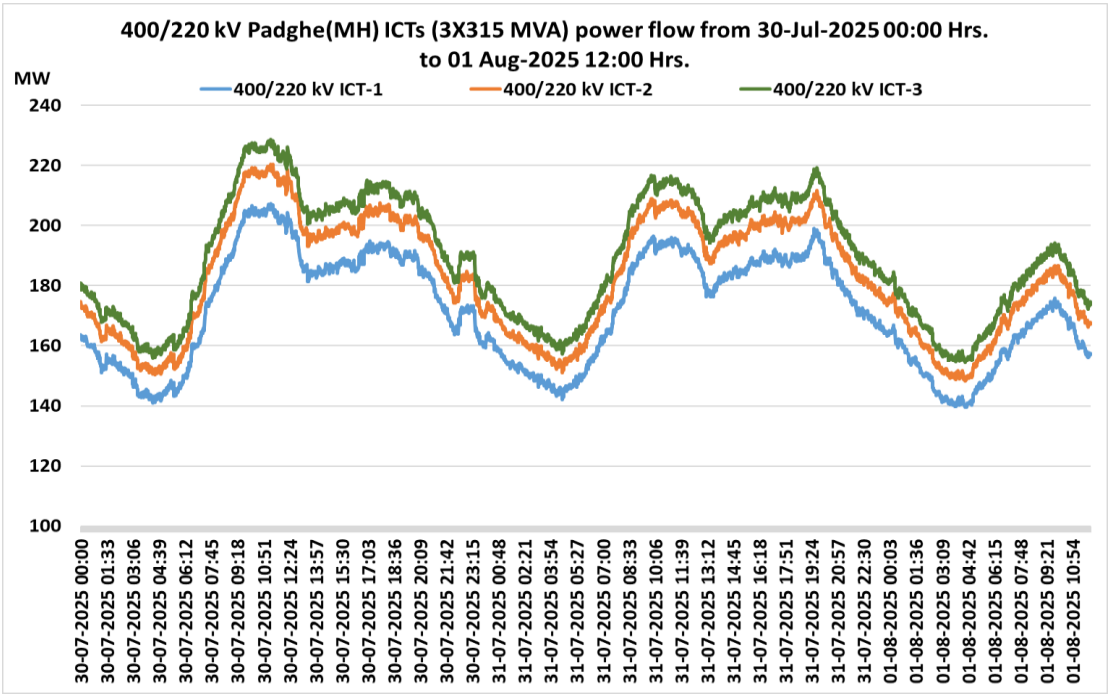


Figure 15: ICT's at Boisar Station.



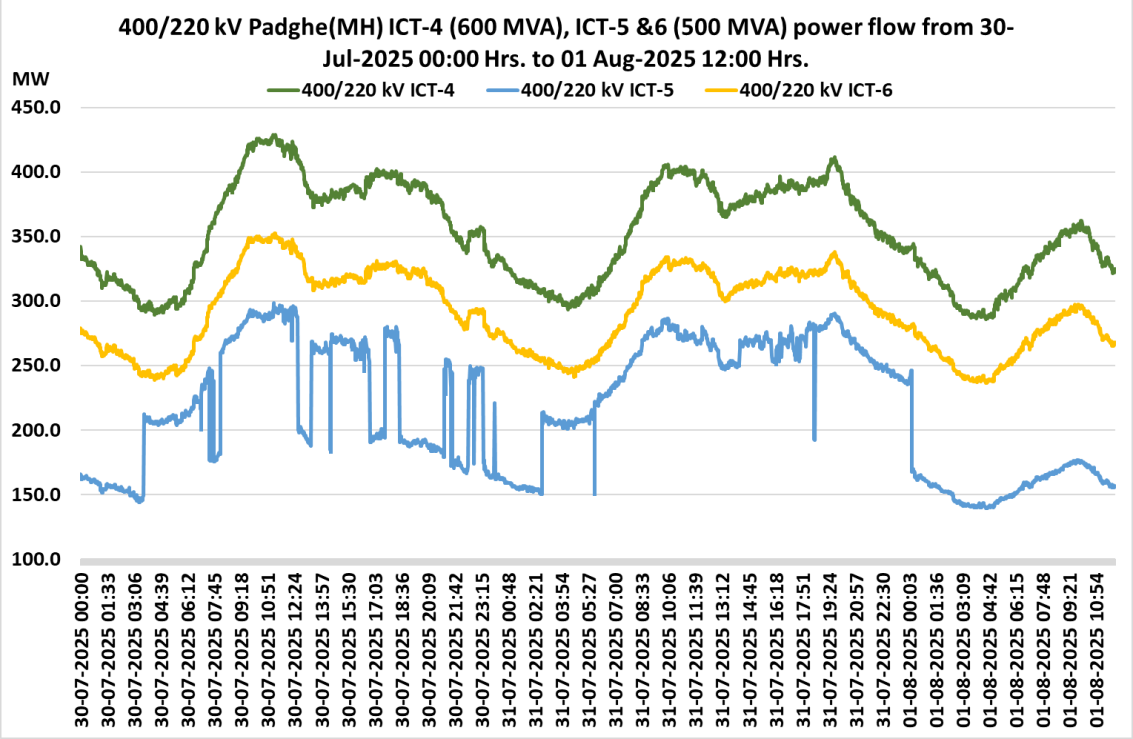


Figure 16: ICT's at Phadge Station.

Note: ICT-5 data is suspect.

Annexure-III

Format for intimating the Failure of Transmission line Towers

1. Date and time of occurrence/discovery of failure
2. Name of Transmission line with voltage level
3. Weather condition on the date of failure:
4. Brief Description of failure:
5. Probable cause of failure:
6. Insulator Failure Details

SN	Type of Insulator	Year of Manufacturing/ Year of Commissioning	Make/ Manufacturer of Insulator	Tower Location	Specific Creepage Distance (kV/ mm)	Details of Electro-mechanical Strength	Configuration of insulator (I / V / Y / tension):	No. of insulators per string and No. of strings per phase

7. Details of Wind Zone & Pollution Level in the areas of above Insulators
8. Original Test Report in support of Quality of the Insulator used
9. Details of any Tests carried out after failure (attach test reports):
10. Post Failure Test Report
11. Quality Assurance Plan followed by the Utility for Insulators
12. OEM Analysis for failure of the Insulators
13. Utility Analysis for failure of the Insulators
14. O&M Plan followed by the Utility
15. Details of last maintenance activity along with date:
16. Voltage & Power flow in the line prior to failure:

Brief on
Unified Real Time Dynamic State Measurement (URTDSM)
Project Phase-II (for ISTS Portion)

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Brief on
Unified Real Time Dynamic State Measurement
URTDSM Project Phase-II
for discussion in RPCs

1.0 Background of URTDSM Phase-II Proposal:

POWERGRID has been entrusted to prepare the DPR for URTDSM Phase-II project in the 13th NPC meeting held on 05.07.2023. The DPR is to be made on basis of 'Report of the Sub-Committee on PMU Placement and Analytics under URTDSM Phase II' constituted by National Power Committee.

Presently 1400 PMUs and 32 Control centres have been established under the URTDSM Phase-I project. This project was implemented with 70% of the project cost as PSDF grant and 30% was through POWERGRID Equity (RTM for 30% portion and no tariff for 70% grant portion).

Similarly, DPR for URTDSM Phase-II Project, which included 4000PMUs(new) and 34 control centres was prepared with funding pattern of 70% PSDF grant and 30% POWERGRID equity in line with Phase-I. The DPR, with an estimate of Rs.3922 Crores, was submitted to NPC/CEA on 11.03.2024. The scope included replacement of 34 control centres and supply of 4000 new PMUs including integration of 1400 existing PMUs.

In the 14th NPC meeting held in Bengaluru on 03.02.2024, DPR status was updated to members. It was suggested to optimize the cost. Further, NPC vide email dated 18.04.2024 has informed that PSDF funding shall not be available for the project and alternate sources of funding shall be explored by POWERGRID. Hence, POWERGRID approached all the Constituents in the RPCs for concurrence for execution of the URTDSM Phase-II Project on RTM basis.

In the 15th NPC Meeting held on 14.11.2024, the committee recommended that "PowerGrid is to submit the revised proposal in consultation with Grid India, only for the existing network after segregating the PMUs and control centers under ISTS and STUs system. The proposal may also be revised to optimize the number of control centers and PMUs at ISTS & STUs system separately. The revised proposal for ISTS portion may be put up to the NCT for further consideration."

During previous NPC & RPC meetings for URTDSM phase-II project, it was discussed that the number of PMUs for URTDSM Phase-II need to be optimized for reducing the overall project cost. The initial DPR prepared by POWERGRID was based on the sub-committee recommended philosophy of PMU placement. However, CEA has published new guidelines for unified philosophy of PMU placement in Indian Grid in March 2025, which will supersede all existing guidelines and sub-committee report etc.,

According to 15th NPC, POWERGRID was entrusted to put up proposal for ISTS portion to NCT and revised proposal for ISTS and STU portion separately for discussion. ISTS portion comprises of Control Centers of Grid-India (NLDCs and RLDCs) and PMUs for Central Sector locations as per latest CEA Guidelines referred above.

POWERGRID prepared DPR for URTDSM Phase-II ISTS portion (upgradation of control centers at NLDCs, RLDCs and installation of new PMUs for Central Sector stations as per latest CEA guidelines). Tentative cost and BOQ for STUs portion (Control centers of SLDCs across India and new PMUs for State sector substations) is also put up for discussion in the 16th NPC.

Upon the concurrence of 16th NPC for proposed DPR for ISTS portion, the same will be put up to NCT for approval in RTM route. For the State Sector, funding methodology is to be finalized in reference to the tentative cost and BOQ.

2.0 Brief Details of URTDSM Phase-I Project:

POWERGRID took up the implementation of URTDSM phase-1 in Jan 2014 and the project was commissioned progressively from 2018 to 2021 with installation of 1409 PMUs and PDCs at 32 control centers, the project was funded 70% from PSDF grant and 30% from POWERGRID equity.

The existing system of URTDSM Phase-I is under maintenance support through Annual Maintenance contract with the contractor, which will end in Jan 2027. The PMUs installed at substations can be in service for fifteen years from their date of commissioning.

3.0 Need for URTDSM Phase-II Project

3.1 Need for new PMUs

After commissioning of URTDSM Phase-I Scheme, expansion of power system has taken place at considerable pace with addition of large-scale renewable generation and incorporation of new transmission system technologies (SVC, STATCOM, FSC, etc.,). Over the years, there have been significant organizational, regulatory, Market operations, and technological changes. The placement of PMUs at new renewable generation plants and other new technology devices (FACTS) is required to know their dynamic response during disturbances and to verify that they are operating under the limits stipulated as per the regulatory norms.

The Committee report on PMU has defined a new philosophy for placement of PMUs and suggested to cover additional equipment such as ICTs, SVCs, Bus Reactors, FSC, TCSC etc. have been considered for measurement through PMU, which were not in Phase-I implementation.

3.2 Upgradation of Control centers

AMC support for the existing control centers will lapse by Jan 2027. Due to this, the maintenance of the existing hardware and software beyond its design life cycle of 7 years will be very challenging owing to lack of spares and services, poor performance and increasing cyber vulnerabilities due to obsolete technologies.

Moreover, the existing hardware and software cannot support the new analytics being envisaged under Phase-II project for better monitoring of the power system.

The URTDSM phase-1 Control centres are having issues of End of Life/End of support due to technical obsolescence of software, hardware, and cyber security appliances and will become technically obsolete. Hence to keep the WAMS URTDSM Control centres functional, the URTDSM phase 2 project which includes the upgrade of existing control centres, is to be implemented on priority.

Increased penetration of Renewable energy has required increase in more monitoring of the regional grids which requires more deployment of PMUs. The URTDSM phase 2 includes installation of PMUs as per latest CEA Guidelines on PMU placement.

3.3 Additional technical factors to establish the need for URTDSM Phase-II System

In addition to the reasons mentioned above, the following are some of the power system aspects which are being handled only with URTDSM system, which is based upon Wide Area Measurement System (WAMS) technology:

(i) PMU measurements provide synchronized voltage, current and phase angle measurements, which is not possible in RTU measurements. The phase angle measurement is valuable information about the state of the grid and the grid operators can take decisions based upon precise measured information instead of thumb rule based upon their experience.

(ii) Based upon the phase angle measurements from both the ends of the line enhanced utilization of Transmission lines closer to thermal loading of line can be achieved.

(iii) In case of system separations, the Synchrophasor measurements display the load angle separation and can be used to determine the change required in generation to synchronize the two isolated systems, this being more accurate data facilitates faster restoration and avoid any jerks or oscillations in the power system.

(iv) One of the most important applications of PMU measurement is post facto analysis so that an incident can be analyzed with high resolution data so the behavior of each element-generator, RE generator, FACTS devices, Power system islands etc.

(v) With increasing grid size and installed capacity from multiple generation sources, the approved placement shall provide a wider footprint of PMU data, increasing observability by the Operator for Grid management.

4.0 DPR for URTDSM Phase-II

As per 15th NPC recommendation, DPR of URTDSM project phase-II (for ISTS portion) in accordance with the latest CEA Guidelines on unified philosophy for PMU placement in Indian Grid is prepared by POWERGRID.

Accordingly, DPR Cost Estimate for URTDSM Phase-II (for ISTS portion) is Rs 1124.35 Cr (including the AMC charges)

POWERGRID proposes to take up URTDSM Phase-II Project (for ISTS portion) on RTM basis (upgradation of all Control Centres for NLDCs and RLDCs of Grid-India, and installation of new PMUs at Central Sector locations as per latest CEA PMU Placement Philosophy), on cost sharing PoC mechanism pan India (**100% RTM route with 70:30 Debt equity ratio**).

5.0 Approach for Implementation for URTDSM Phase II Project

The approach for Implementation for URTDSM phase-II project is as follows:

(i) The URTDSM Phase-II project shall be implemented initially for ISTS portion covering NLDCs (2), RLDCs (5), new PMUs at Central Sector locations and remote consoles for all SLDCs.

(ii) Phase-II for STU portion (SLDCs and State sector/IPP stations) shall be implemented separately, subject to finalization of funding method for state portion.

6.0 Scope of Work for URTDSM Phase II Project:

The scope of work under URTDSM Phase-II project is proposed as follows.

1. Installation of approximately 1070 nos. of PMUs at the Central Sector Substations and Power plants across the country as per the lates CEA Guidelines for unified PMU placement philosophy in Indian Grid.
2. Replacement/Establishment of PDCs and associated control center equipment comprising of software and hardware at 7 Control centers.
    a. Main and Backup NLDCs (2 nos.)
    b. All RLDCs (5 nos.)
    c. Remote Consoles for CEA/RPCs/SLDCs - 31 nos
3. The hardware and software to be installed at Control Centers shall be sized to accommodate all the PMUs currently installed under Phase-I, and the proposed PMUs under Phase-II with provision for future expansion of about 100% over and above.
4. The FO based communication system existing and being established by POWERGRID and Constituents shall meet the requirement of Phase-II. *The addition/augmentation of communication link is NOT envisaged under the scope of URTDSM Phase-II project DPR*.

5. **New Analytical applications**: Analytical applications have been suggested by the subcommittee for deployment under Phase-II scheme. Few of them, like Linear State Estimator and Oscillation Monitoring Application, are already deployed in Phase-I. However, due to obsolescence of the hardware/software issues in the existing control centers, they need to be replaced with new software and hardware. The following are the new analytical applications being proposed under Phase-II at all 7 control centers:

    a. Linear State Estimator
    b. Oscillation Monitoring Application.
    c. Real time automated event detection along with early warning system and ROCOF calculation over variable window.
    d. Voltage Stability analytics (VSA)
    e. WAMS based contingency analysis, security assessment & Islanding Detection.
    f. Real time Inertia Estimation and monitoring.
    g. Post-mortem analytics.
    h. Generator Model Validation
    i. Wide Area Control Systems (at selected nodes only)
        (i) WAMS based automatic load shedding (AUFLS and df/dt):
        (ii) Control of HVDC, PSS and STATCOM for damping system oscillations
    j. Response of Wind Farm and Solar PV for LVRT, Reactive Power etc.

    The above analytical applications are the requirements suggested based on the feedback of the system operator. It will be explored to procure them either from the prospective Bidders/Software OEMs or to develop them in association with premier academic institutes.

6. Integration of the existing PMUs.

7. Integration of the Phase-II PDCs with the existing SCADA/EMS systems at each location.

8. Integration with existing Video Projection System (VPS) (supplied under separate project) at each location.

9. Consultancy Services from premier academic institutions for development of new analytics under phase-II implementation, if required.

10. Engagement of Consultant for design of large scale WAMS system comprising of IT infrastructure for handling the vast amount of PMU data in real-time and historical data.

11. Capacity building through training of engineers from Grid-India (for Control center portion) & Central Sector Utilities (for PMU portion) are proposed as part of this URTDSM Phase-II project.

In addition to the Supply and Services under Phase-II, it is also proposed to include the Annual Maintenance Contract (AMC) charges through 100% RTM basis (70% debt and 30% Equity from POWERGRID), similar to the AMC contract being operated by POWERGRID for existing Phase-I system.

6.1 Location of SLDCs for PDC replacement and no of PMUs in each Region under URTDSM Phase II Project (ISTS portion):

Region→	NR	ER	NER	WR	SR
No of Control Centers	2	2	1	1	1
Location of 7 control centers	1. Main NLDC 2. NRLDC	1. Backup NLDC 2. ERLDC	1. NERLDC	1. WRLDC	1. SRLDC
PMUs (1070 No's) in Central Sector locations of each Region (as per latest CEA Guidelines)	218	303	151	309	89

7.0 URTDSM Phase II Project DPR Cost for ISTS portion across India:

BoQ Category	Software	Hardware	Auxiliary Power System	Services	Training	AMC	Total (including T&D)	T&D
PDCs at NLDC	144.03	21.26	2.06	2.76	0.00	3.34	173.46	25.62
PDCs at RLDC	348.57	53.9423	4.00	6.90	7.43	8.35	429.19	63.44
PMUs	17.568	321.1894	0.00	0.00	2.11	20.11	360.98	53.37
Mandatory Spares	0.00	17.1105	0.68	0.00	0.00	0.00	17.79	2.63
Total	**510.17**	**413.50**	**6.74**	**9.66**	**9.54**	**31.80**	**981.42**	**145.05**

DPR is based on a 100% RTM basis with 70% debt and 30% equity from POWERGRID.

Sl. No.	Description	Total Cost (Rs. in crores)
1	Unified Real Time Dynamic State Measurement Phase-II *(Supply, services, taxes & duties, testing & commissioning)*	949.62
2	Overheads *(IEDC, Contingencies)*	75.97
3	Consultancy charges for Panel of Experts	10.00
	Sub-Total	1035.59
4	Interest during Construction (IDC)	56.96
	TOTAL (Supply, Services)	1092.55
5	AMC charges and other services during AMC	**31.80**
	GRAND TOTAL including AMC	**1124.35**

The abstract cost estimate is given at ***Annexure***
The above project cost including AMC is based on the Budgetary quote from M/s L&T, the actual project and AMC charges will be known after the process of tendering and award, subject to approval of this proposal by Regulatory Authority.

7.1 Target Beneficiaries

The beneficiaries of the project would be all the designated ISTS customers (DICs) *as per POC mechanism for the URTDSM Phase-II (ISTS portion) as per CERC regulations.* ***(as per Clause no. 78 of CERC Tariff Regulations 2024-29).***

7.2 Project Strategy

The Unified Real Time Dynamic State Measurement System (URTDSM) Phase-II Project shall be implemented by POWERGRID. The PMU installation shall be taken up at those locations where Fiber Optic based communication system is either existing or is being installed under various projects, which shall be available by Dec' 2027. The tariff on the investments for the same shall be recovered from the beneficiaries (SLDCs) as per CERC regulations.

Upon the concurrence of POWERGRID's RTM proposal for ISTS portion by NPC/NCT, POWERGRID shall proceed for further tendering process.

The project shall be planned for a lifetime of 7 years for the IT infrastructure at Control Centers. The AMC shall be taken for 1-year (Defect liability period) and 6 years (AMC period) after operational acceptance (with an additional provision to extend the AMC by another 2 years on the same terms and conditions.

8.0 O&M facilities

The URTDSM system shall be used by respective NLDC/ RLDC/ Central Sector substations and the maintenance responsibility for the subject project shall be under the scope of Grid-India (for all 7 Control centers) and by respective Central Sector Utility for the PMU locations.

After the defect liability period (DLP) of 1 year, 6-year maintenance support by OEM shall be kept in the Project.

POWERGRID hereby proposes for funding the AMC charges also along with Supply and Services charges for SLDCs portion through 100% RTM route under this URTDSM Phase-II scheme. However, The charges for O&M of the system shall be finalized as per CERC/ RPC approval.

9.0 Time Frame

The subject project is scheduled to be commissioned within **36 months** from the date of Investment Approval.

10.0 Conclusion:

The URTDSM phase-1 Control centres are nearing their Operational life and will become technically obsolete. Hence to keep the WAMS URTDSM Control centres functional, the URTDSM Phase-II project which includes an upgrade of existing control centres, is to be implemented on priority.

Increased penetration of Renewable energy has required increase in more monitoring of the regional grids which requires more deployment of PMUs. The proposal for URTDSM Phase-II (ISTS portion) includes installation of PMUs at Central Sector locations as per the latest CEA guidelines.

1. *"**POWERGRID proposes to take up URTDSM Phase-II Project (for ISTS portion) on pan India basis, on cost sharing mechanism (100% RTM route with 70:30 Debt Equity ratio)**".*

2. ***Deliberation on URTDSM Phase-II for STU portion, subject to finalization of funding methodology for States.***

   Put up for deliberation in NPC/RPCs.

URTDSM Phase-II WAMS System - BoQ for Main & Backup NLDC				
Sr.No	Name of the item	Unit	Main NLDC Qty	Backup NLDC Qty
A	SOFTWARE			
1	PDC Software	Lot	1	1
2	WAMS Visualisation (UI) Software	Lot	1	1
3	Analytical Applications			
(a)	Linear State Estimator	Lot	1	1
(b)	Oscillation Monitoring Application	Lot	1	1
(c)	Real time automated event detection along with early warning system and ROCOF calculation over variable window	Lot	1	1
(d)	WAMS based Voltage Stability analytics (VSA)	Lot	1	1
(e)	WAMS based contingency analysis, security assessment & Islanding Detection	Lot	1	1
(f)	Generator Model Validation, Real time Inertia Estimation and monitoring	Lot	1	1
(g)	Post-mortem analytics	Lot	1	1
(h)	Wide Area Control Systems			
i.	WAMS based automatic load shedding (AUFLS and df/dt)	Lot	1	1
ii.	Control of HVDC, PSS and STATCOM for damping system oscillations	Lot	1	1
(i)	Response of Wind Farm and Solar PV for LVRT, Reactive Power.	Lot	1	1
4	Programming Development System (PDS) Software	Lot	1	1
5	Commercial Off-The-Shelf (COTS) Softwares			
(a)	Data Historian Software	Lot	1	1
(b)	Identity Management Software	Lot	1	1
(c)	Network Access Control (NAC)	Lot	1	1
(d)	Patch Management Software	Lot	1	1
(e)	Virtualisation Software for all the virtual servers envisaged and required under the project along-with centralised management software	Lot	1	1
(f)	Operating System for all the servers	Lot	1	1
(g)	Host based intrusion prevention system (HIPS) with centralised management	Lot	1	1
(h)	End Point Security Solution	Lot	1	1
(i)	Centralised Management and Log Analyser of all FWs	Lot	1	1
(j)	Network Management System (NMS)	Lot	1	1
(k)	SIEM (Security Information and Event management)	Lot	1	1
(l)	VAPT Tool	Lot	1	1
6	Report Development & Generation Software	Lot	1	1
7	Storage system			
(a)	SAN Software	Lot	1	1

(b)	NAS Software	Lot	1	1
8	SMS & Email Interface	Lot	1	1
9	Backup Solution Software	Lot	1	1
10	Data exchange Software with external applications	Lot	1	1
B	**HARDWARE**			
1	PDC Server sized for data of 10000 PMUs	No.	2	2
2	Analytical Applications Servers	No.	2	2
3	Historian Servers	No.	2	2
(a)	Data Historian Server			
(b)	Report Development & Generation Server			
4	WAMS Visualisation (UI) Server	No	2	2
5	Programming Development System (PDS) Server	No	1	0
6	Management Applications Servers			
(a)	Identity Management Server	No.	2	2
(b)	NMS Server			
(c)	Network Access Control (NAC) Server	No.	2	2
(d)	End Point Security Solution			
(e)	Centralised Management & Log analyser of Firewall (Internal)			
7	Internal DMZ Servers			
(a)	SIEM Server	No	2	2
(b)	Centralised Management & Log analyser of Firewall (External)			
8	External DMZ Servers			
(a)	End Point Security Solution	No	2	2
(b)	Patch Management Server			
9	Anti APT	No	1	1
10	Storage System			
(a)	Storage Solution of Minimum 1200TB, RAID10, SAN or equivalent along with SAN Management Server	No.	1	1
(b)	NAS Storage of 100TB	No.	1	1
11	Workstation consoles with 23.8" dual monitor			
(a)	PDC application	No.	2	2
(b)	PDS Application	No.	1	1
(c)	Analytical Applications	No.	1	1
(d)	External DMZ	No.	1	1
(e)	Inernal DMZ	No.	1	1
(f)	Management Applications	No.	1	1
(g)	Historian Applications	No.	1	1
(h)	Server Management Console	No	1	1
(i)	Remote Diagnostic Console	No	1	1
12	Color Laser Printer	No.	1	1
13	Time System (NavIC & GPS based)	Lot	2	2
14	Firewalls (with Minimum 10Gbps NGTP)			
(a)	External Firewall with NIPS for Internet and Remote consoles	No.	2	2
(b)	External Firewall with NIPS for PDC/PMU/SCADA data	No.	2	2
(c)	Internal Firewall with NIPS	No.	2	2
15	WAN Router cum Firewall			

(a)	For Communicating with PDCs at RLDCs (8 x Fiber Optic Ports, 8 Ethernet ports (1Gbps))	No.	2	2
(b)	For communicatiing with Remote Consoles (8 x Fiber Optic ports, 8 Ethernet ports (1Gbps))	No.	2	2
(c)	For Internet/Corporate Intranet connectivity (8 x Fiber Optic ports, 8 Ethernet ports (1Gbps))	No.	2	2
(d)	WAN router at Remote Console end (4 Ethernet ports (1Gbps))	No.	3	
16	L3 LAN Switches (10G FO ports) for the following LANs			
(a)	48-port L3 LAN Switch for PDC LAN	No.	2	2
(b)	24 port L3 LAN Switch for Historian LAN	No.	2	2
(c)	24 port L3 LAN Switch for External DMZ LAN	No.	2	2
(d)	24 port L3 LAN Switch for Internal DMZ LAN	No.	2	2
(e)	24 port L3 LAN Switch for Management LAN	No.	2	2
(f)	24 port L3 LAN Switch for Analytics LAN	No.	2	2
(g)	32 port FC Switch for SAN system	No.	2	2
17	Remote Consoles, equivalent to Work station console mentioned at Item No.4 above (for NTAMC and other Strategic locations)	No.	7	
18	Server for Backup Solution	No.	1	1
19	Backup Appliance	No.	1	1
20	Laptop for VAPT	No	1	1
21	Server for Data Exchange with external applications	No	1	1
22	Any other additional hardware at Control center end for implementing Wide Area Measurement Protection and Control (WAMPAC) system	No	1	1
C	**Auxiliary Power System**			
(a)	120 kVA (96kW at 0.8 pf) UPS running in parallel	No.	2	2
(b)	VRLA type Battery banks for above UPS (each bank of 230.4 kVAH)	No.	2	2
(c)	Input ACDB (600kVA rating)	No.	1	1
(d)	Output ACDB (400kVA rating)	No.	1	1
(e)	Accessories for maintenance of VRLA type batteries	Lot	1	1
(f)	Power Distribution and cabling work required to establish UPS	Lot	1	1
(g)	UPS Monitoring System and it's integration with URTDSM System	Lot	1	1
D	**SERVICES**			
1	Integration of WAMS system with following units/applications:			
(a)	Integration with the SCADA/EMS Sytem at NLDC and backup NLDC respectively	No.	1	1
(b)	Integration with 3rd party applications	Lot	1	1
(c)	Integration (at Control center end) of existing PMUs	No	3,000	3,000
(d)	Integration (at control center end) of new PMUs supplied under this project	No	1,100	1,100
(e)	Integration of WAMS System with existing Video Projection System (VPS) of SCADA/EMS System in respective control center	Lot	1	1
(f)	Cyber Security Audit by Cert-IN certified Auditors during FAT.	Lot	1	1
(g)	Cyber Security Audit by Cert-IN certified Auditors during SAT.	Lot	1	1
(h)	Dismantling and Buyback of existing WAMS system of URTDSM Phase-I (after successful parallel operation)	Lot	1	1
(i)	SMS integration with service provider Email integration with owner email system	Lot	1	1

E	**Training**			
1	Training - Man days @15 Days x 18 persons	Man-days	270	
F	**Annual Maintenance Contract**			
(a)	Annual maintenance contract of WAMS System and all the equipment supplied in the project for a period of 7 years (1 year DLP and 6 years AMC)	Lot	1	1
(b)	Annual Training under AMC period	Lot	1	1
(c)	Six Monthly Cyber Security Audit by Cert-IN certified Auditors during 7 years AMC period	Lot	1	1
(d)	Patch Management including Signature updates for all Cyber security equipments for 7 years	Lot	1	1
(e)	Integration of new PMUs data from PDCs at respective RLDCs with PDC at NLDC during entire AMC period	No	5,000	5,000

>> The sizing for Historian Storage at NLDCs is considered @25 samples/second reporting rate of the existing and new PMUs including 100% expansion

URTDSM Phase-II WAMS System - BoQ for 5 RLDCs								
Sr.No	Name of the item	Unit	Name of RLDC					Total Qty
			NRLDC	ERLDC	NERLDC	SRLDC	WRLDC	
A	**SOFTWARE**							
1	PDC Software	Lot	1	1	1	1	1	5
2	WAMS Visualisation (UI) Software	Lot	1	1	1	1	1	5
3	Analytical Applications							
(a)	Linear State Estimator	Lot	1	1	1	1	1	5
(b)	Oscillation Monitoring Application	Lot	1	1	1	1	1	5
(c)	Real time automated event detection along with early warning system and ROCOF calculation over variable window	Lot	1	1	1	1	1	5
(d)	WAMS based Voltage Stability analytics (VSA)	Lot	1	1	1	1	1	5
(e)	WAMS based contingency analysis, security assessment & Islanding Detection	Lot	1	1	1	1	1	5
(f)	Generator Model Validation, Real time Inertia Estimation and monitoring	Lot	1	1	1	1	1	5
(g)	Post-mortem analytics	Lot	1	1	1	1	1	5
(h)	Wide Area Control Systems							
i.	WAMS based automatic load shedding (AUFLS and df/dt)	Lot	1	1	1	1	1	5
ii.	Control of HVDC, PSS and STATCOM for damping system oscillations	Lot	1	1	1	1	1	5
(i)	Response of Wind Farm and Solar PV for LVRT, Reactive Power.	Lot	1	1	1	1	1	5
4	Programming Development System (PDS) Software	Lot	1	1	1	1	1	5
5	Commercial Off-The-Shelf (COTS) Softwares							
(a)	Data Historian Software	Lot	1	1	1	1	1	5
(b)	Identity Management Software	Lot	1	1	1	1	1	5
(c)	Network Access Control (NAC)	Lot	1	1	1	1	1	5
(d)	Patch Management Software	Lot	1	1	1	1	1	5
(e)	Virtualisation Software for all the virtual servers envisaged and required under the project along-with centralised management software	Lot	1	1	1	1	1	5
(f)	Operating System for all the servers	Lot	1	1	1	1	1	5
(g)	Host based intrusion prevention system (HIPS) with centralised management	Lot	1	1	1	1	1	5
(h)	End Point Security Solution with centralised management	Lot	1	1	1	1	1	5
(i)	Centralised Management and Log Analyser of all FWs	Lot	1	1	1	1	1	5
(j)	Network Management System	Lot	1	1	1	1	1	5
(k)	SIEM (Security Information and Event management)	Lot	1	1	1	1	1	5
(l)	VAPT Tool	Lot	1	1	1	1	1	5
6	Report Development & Generation Software	Lot	1	1	1	1	1	5
7	Storage system							
(a)	SAN Software	Lot	1	1	1	1	1	5
(b)	NAS Software	Lot	1	1	1	1	1	5
8	SMS & Email Interface	Lot	1	1	1	1	1	5
9	Backup Solution Software	Lot	1	1	1	1	1	5
10	Data exchange Software with external applications	Lot	1	1	1	1	1	5
B	**HARDWARE**							
1	PDC Server sized for data of 2500 PMUs	No.	2	2	0	2	2	8
2	PDC Server sized for data of 1000 PMUs	No.	0	0	2	0	0	2
3	Analytical Applications Servers	No.	2	2	2	2	2	10
4	Historian Servers	No.	2	2	2	2	2	10

(a)	Data Historian Server							
(b)	Report Development & Generation Server							
5	WAMS Visualisation (UI) Server	No	2	2	2	2	2	10
6	Programming Development System (PDS) Server	No	1	1	1	1	1	5
7	Management Applications Servers							
(a)	Identity Management Server	No.	2	2	2	2	2	10
(b)	NMS Server							
(c)	Centralised Management & Log analyser of Firewall (Internal)	No.	2	2	2	2	2	10
(d)	Network Access Control (NAC) Server							
(e)	End Point Security Solution							
8	Internal DMZ Servers	No	2	2	2	2	2	10
(a)	SIEM Server							
(b)	Centralised Management & Log analyser of Firewall (External)							
9	External DMZ Servers	No	2	2	2	2	2	10
(a)	End Point Security Solution							
(b)	Patch Management Server							
10	Anti APT	No	1	1	1	1	1	5
11	Storage System							
(a)	Storage Solution of Minimum 600TB, RAID10, SAN or equivalent along with SAN Management Server	No.	1	1	0	1	1	4
(b)	Storage Solution of Minimum 250TB, RAID10, SAN or equivalent along with SAN Management Server	No.	0	0	1	0	0	1
(c)	NAS Storage of 50TB	No.	1	1	0	1	1	4
(d)	NAS Storage of 20TB	No.	0	0	1	0	0	1
12	Workstation consoles with dual monitor							
(a)	PDC application	No.	2	2	2	2	2	10
(b)	PDS Application	No.	1	1	1	1	1	5
(c)	Analytical Applications	No.	1	1	1	1	1	5
(d)	External DMZ	No.	1	1	1	1	1	5
(e)	Inernal DMZ	No.	1	1	1	1	1	5
(f)	Management Applications	No.	1	1	1	1	1	5
(g)	Historian Applications	No.	1	1	1	1	1	5
(h)	Server Management Console	No	1	1	1	1	1	5
(i)	Remote Diagnostic Console	No	1	1	1	1	1	5
13	Color Laser Printer	No.	1	1	1	1	1	5
14	Time System (NavIC & GPS based)	Lot	2	2	2	2	2	10
15	Firewalls (With Minimum 8Gbps NGTP)							
(a)	External Firewall with NIPS for Internet and Remote consoles	No.	2	2	2	2	2	10
(b)	External Firewall with NIPS for PDC/PMU/SCADA data	No.	2	2	2	2	2	10
(c)	Internal Firewall with NIPS	No.	2	2	2	2	2	10
16	WAN Router cum Firewall							
(a)	For Communicating with PDCs at NLDCs (8 x Fiber Optic Ports, 8 Ethernet ports (1Gbps))	No.	2	2	2	2	2	10
(b)	For Communicating with PDCs at SLDCs (8 x Fiber Optic Ports, 8 Ethernet ports (1Gbps))	No.	2	2	2	2	2	10
(c)	For communicatiing with Remote Consoles (8 x Fiber Optic ports, 8 Ethernet ports (1Gbps))	No.	2	2	2	2	2	10
(d)	For Internet/Corporate Intranet connectivity (8 x Fiber Optic ports, 8 Ethernet ports (1Gbps))	No.	2	2	2	2	2	10
(e)	WAN router at Remote Console end (4 Ethernet ports (1Gbps))	No.	2	2	2	2	2	10

17	L3 LAN Switches (10G FO ports) for the following LANs							
(a)	48-port L3 LAN Switch for PDC LAN	No.	2	2	2	2	2	10
(b)	24 port L3 LAN Switch for Historian LAN	No.	2	2	2	2	2	10
(c)	24 port L3 LAN Switch for External DMZ LAN	No.	2	2	2	2	2	10
(d)	24 port L3 LAN Switch for Internal DMZ LAN	No.	2	2	2	2	2	10
(e)	24 port L3 LAN Switch for Management LAN	No.	2	2	2	2	2	10
(f)	24 port L3 LAN Switch for Analytics LAN	No.	2	2	2	2	2	10
(g)	32 port FC Switch for SAN system	No.	2	2	2	2	2	10
18	Remote Consoles, equivalent to Work station console mentioned at Item No.4 above	No.			10			10
19	Server for Backup Solution	No.	1	1	1	1	1	5
20	Backup Appliance	No.	1	1	1	1	1	5
21	Laptop for VAPT	No	1	1	1	1	1	5
22	Server for Data Exchange with external applications	No	1	1	1	1	1	5
23	Any other additional hardware at Control center end for implementing Wide Area Measurement Protection and Control (WAMPAC) system	No	1	1	1	1	1	5
C	**Auxiliary Power System**							
(a)	60 kVA (48kW at 0.8 pf) UPS running in parallel	No.	2	2	2	2	2	10
(b)	VRLA type Battery banks for above UPS (each bank of 115.2 kVAH)	No.	2	2	2	2	2	10
(c)	Input ACDB (450kVA rating)	No.	1	1	1	1	1	5
(d)	Output ACDB (300kVA rating)	No.	1	1	1	1	1	5
(e)	Accessories for maintenance of VRLA type batteries	Lot	1	1	1	1	1	5
(f)	Power Distribution and cabling work required to establish UPS	Lot	1	1	1	1	1	5
(g)	UPS Monitoring System and it's integration with URTDSM System	Lot	1	1	1	1	1	5
D	**SERVICES**							
1	Integration of WAMS system with following units/applications:							
(a)	Integration with the SCADA/EMS Sytem at respective RLDC	No.	1	1	1	1	1	5
(b)	Integration with 3rd party applications	Lot	1	1	1	1	1	5
(c)	Integration (at Control center end) of existing PMUs	No.			3,000			3000
(d)	Integration (at Control center end) of new PMUs supplied under this project	No.	218	303	151	89	309	1070
(e)	Integration of WAMS System with existing Video Projection System (VPS) of SCADA/EMS System in respective region	Lot	1	1	1	1	1	5
(f)	Cyber Security Audit by Cert-IN certified Auditors during FAT.	Lot	1	1	1	1	1	5
(g)	Cyber Security Audit by Cert-IN certified Auditors during SAT.	Lot	1	1	1	1	1	5
(h)	Dismantling and Buyback of existing WAMS system of URTDSM Phase-I (after successful parallel operation)	Lot	1	1	1	1	1	5
(i)	SMS integration with service provider Email integration with owner email system	Lot	1	1	1	1	1	5
E	**Training**							
1	Training - Man days @15 Days x 24 persons in each Region	Man-days	360	360	360	360	360	1,800
F	**Annual Maintenance Contract**							
(a)	Annual maintenance contract of WAMS System and all the equipment supplied in the project for a period of 7 years (1 year DLP and 6 years AMC)	Lot	1	1	1	1	1	5

(b)	Annual Training under AMC period	Lot	1	1	1	1	1	5
(c)	Six Monthly Cyber Security Audit by Cert-IN certified Auditors during 7 years AMC period	Lot	1	1	1	1	1	5
(d)	Patch Management including Signature updates for all Cyber security equipments for 7 years	Lot	1	1	1	1	1	5
(e)	Integration of new PMUs (from existing substations or from new substations) with PDCs at RLDCs during entire AMC period	No	1,200	800	700	1,100	1,200	5000
(f)	Integration of PDCs at SLDCs with PDCs at respective RLDCs during AMC period	No	9	5	3	5	4	26

>> The sizing for Historian Storage at RLDCs is considered @50 samples/second reporting rate of the existing and new PMUs including 100% expansion

BOQ for Central Sector PMUs to be procured under URTDSM Phase-II								
S.No	Name of the item	Unit	Regionwise Qty					Total Qty
			NR	ER	NER	SR	WR	
A	**SOFTWARE**							
1	PMU configuration software	Lot	12	6	4	6	4	32
B	**HARDWARE**							
1	PMUs		218	303	151	89	309	1070
2	Panel for mounting PMUs (complete with all necessary accessories, cables etc. as per specification) along with identified analog channels / modules	Lot	1	1	1	1	1	5
3	Time System (GPS receiver)	Lot	218	303	151	89	309	1070
4	Substation Grade Layer-3 LAN Switches with 10 ports minimum i.1 Gbps Fibre port- 4 nos. ii.1 Gbps Cu ports- 2 nos. iii.100 Mbps Cu ports- 4 nos	No.	135	170	55	75	210	645
5	Substation Grade Layer-2 LAN Switches with 10 ports minimum i.1 Gbps Fibre port- 4 nos. ii.100Mbps Fibre ports- 4 nos. iii.100Mbps Cu ports – 2 nos.	No.	109	152	76	45	155	535
6	Armored Fibre Optic Cable and associated termination	Lot	135	170	55	75	210	645
7	LIU - FO PATCH PANEL-12 PORT	No.	244	322	131	120	365	1180
8	PMU configuration tool (Laptop)	No.	12	6	4	6	4	32
9	Integration (at substation end) of new PMUs supplied under this project with PDCs of respective Control center	No.	218	303	151	89	309	1070
B	**TRAINING**							
1	Training (For All Central Sector locations in each Region, 3 days for 5 persons)	Man days	150	90	90	90	90	510
C	**Annual Maintenance Contract**							
(a)	Annual maintenance contract of PMUs and all the associated equipment supplied in the project for a period of 7 years (1 year DLP and 6 years AMC)	Lot	1	1	1	1	1	5

>> The PMU quantity mentioned above is indicative requirement under Phase-II for only Central Sector portion.
>> Each PMU shall be supporting measurement of 2sets of Voltage and Current phasors (i.e, of 2 elements) as a minimum and shall comply with latest version of IEEE C37.118.2 and IS/IEC 60255-118.1standards with latest amendments.

URTDSM Phase-II - BOQ for Mandatory Spares			
Sl.no	**Item description**	**Unit**	**Total Qty**
A	**Spares for URTDSM system**		
1	Servers one of each type at every RLDC & NLDC	Lot	1
2	**Storage System**		
(i)	10% of the critical items at every control center like Dual redundant power supplies, controllers and storage disks/ specialized storage etc.	Lot	1
3	Workstation console with dualcolour monitor (@10% of Total Supply in each Region/State)	Lot	1
4	LAN switch one of each type at every RLDC/NLDC	Lot	1
5	Internal Firewall at every RLDC and NLDC	Lot	1
6	External Firewall at every RLDC and NLDC	Lot	1
7	WAN Routers one of each type at every RLDC and NLDC	Lot	1
8	Time System (GPS receiver) (@10% of Supply)	Lot	1
9	PMU (complete with all necessary accessories, cables etc. as per specification) along with additional analog channels / modules	No.	110
B	**Spares for Auxiliary Power Supply system**		
1	MCCB/MCB/Isolator/ Switch/Contactor of each type & rating (as applicable & used inside UPS panel)	Lot	7
2	Fuse of each type & rating (if applicable)	Lot	35
3	DC Filter assembly	Lot	7
4	Input AC Filter assembly	Lot	7
5	Output AC Filter assembly	Lot	7
6	Electronic Printed Circuit Board / Card of each type (including all cards/modules for rectifier/charger, inverter, system card, display module, interface cards etc.)	Lot	7
7	Power Semiconductor devices of each type & rating such as SCRs, IGBTs etc. for rectifier/charger module, Inverter module, Static Switch module for all the three phases (exclude those items which are covered under item-6 above)	Lot	7

DATA CLASSIFICATION : RESTRICTED

ANNEXURE - 4.0

ABSTRACT COST ESTIMATE

(BASE COST)

Unified Real Time Dynamic State Measurement (URTDSM)
Project Phase-II

DPR Cost Estimate (Based on budgetary offers)

Sl. No.	DESCRIPTION		AMOUNT (Rupees in Crs.)
	Equipment Cost		
A	**Supply**		
	Sub- Total A		949.62
B	**Services**		
i	Installation, Testing and Commissioning		
ii	Training charges		
	Sub- Total B (i to ii)		**Included above in A**
C	**Taxes and Duties**		
i	GST @ 18%, 28% (as applicable on Supply cost & Services) on **(A+B)**		**Included above in A**
D	Inland Freight and Insurance (as applicable on **A**)		**Included above in A**
E	**Sub Total A TO D**		**949.62**
F	Incidental Expediture During construction (IEDC)	**@ 5% of [F]**	47.48
G	Contingencies	**@ 3% of [F]**	28.49
H	Consultancy charges for URTDSM Phase-II for Panel of Experts		10.00
I	**Sub Total (A TO H)**		**1,035.59**
J	Interest During Construction (IDC)		56.96
K	**GRAND TOTAL**		**1,092.55**
L	Maintenance charges for 1 year during warranty period and 6 years after warranty period excl. GST* (AMC) incl. Cyber Security Audit		**26.95**
	GST @ 18% on AMC		**4.85**

Note:
1. Debt:Equity ratio has been considered as 70:30.
2. Interest rate on Loan has been considered @ 11.00% for Domestic Loan subject to actuals.
3. The project is scheduled to be commissioned by 01.09.2028. However, for the purpose of phasing of funds, Investment approval has been assumed on 01.09.2025.
4. IDC has not been calculated on AMC. Annual Maintenance Charges is not part of the Project cost. However, the same shall be part of bidding documents and to be borne out of O&M expenditure